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On general relativity:

how to physics in a curved spacetime.

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Abstract

In these notes, we will study general relativity.

Part I

Einstein field equations

Chapter 1

Principles of general relativity

The theory of general relativity was formulated by Einstein i

In this chapter, we will briefly review all the underline principles that led Einstein to formulate its theory: principle of general relativity, principles of equivalence and principle of general covariance.

1.1 Principle of general relativity

All physics measurements are local, since they are performed by apparati in a finite spatial and temporal extension. Therefore, by means of differential geometry, we need to write equations that take the same form in any arbitrary (local) reference frame and they must be independent of that.

Principle 1.1 (of general relativity)
The laws of physics are the same in all reference frame.

This implies that physical laws must be written in terms of tensors. Observers are physical apparati that move along trajectories in spacetime and we can cover a portion of spacetime with a reference frame starting from that.

1.2 Principles of equivalence

All fundamental forces are described by charges of opposite sign, therefore they can be neglected using zero charge objects. However, we cannot do that with gravity since it is always attractive. The only way to eliminate gravity is considering a free-falling observer.

Principle 1.2 (of equivalence (weak))
The gravitational mass m_g is equal to the inertial mass m_i.

Principle 1.3 (of equivalence (Einstein))
Motion in a uniform gravitational field is equal to free fall.

In other words, we cannot distinguish by a local experiment, if we are in a uniform gravitation field or if we are free-falling.

Principle 1.4 (of equivalence (strong))
There always exists a local reference frame in which all gravitational effects vanish.

This means that inertial frames are free-falling reference frames and they can be defined only in a small neighbourhood of a point. In fact, for these frames, the metric reduces to Minkowski and we can use Killing vectors tangent to these trajectories to build a local reference frame in each point, which is a Gaussian normal frame.

1.3 Principle of general covariance

Consequently, for inertial frames, we have $\pounds_\mu \simeq \partial_\mu \simeq \nabla_\mu$ and we restore special relativity.

Principle 1.5 (of general covariance)
The laws of physics in a general reference frame are obtained by special relativity by replacing differential geometry quantities (metric, indices, covariant derivatives).

This can be made because of the local flatness theorem.

Chapter 2

Einstein field equations

In this chapter, we will study what are test particles and what are the sources of gravity. Furthermore, we will derive the Einstein field equation.ì by physics analogy.

2.1 Test particles

Consider a massive test particle of 4-velocity $u^\mu = dx^\mu/d\tau$, which is not subjected to any force, in a free-falling frame. Therefore, we can use special relativity and write down the relativistic Newton's law

$$\frac{d^2x^\alpha}{d\tau^2} = 0 \ ,$$

where $t = \gamma\tau$ is the proper time. Moreover, in a local inertial frame $\Gamma^\alpha_{\mu\nu} \sim g_{\mu\nu,\beta} \sim \eta_{\mu\nu,\beta} = 0$, hence

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0 \ .$$

This is exactly the geodesic equation

$$u^\mu \nabla_\mu u^\alpha = 0 \ .$$

Proof. In fact,

$$u^\mu \nabla_\mu u^\alpha = \underbrace{u^\mu \partial_\mu}_{d/d\tau} u^\alpha + \Gamma^\alpha_{\mu\nu} u^\mu u^\nu = \frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0 \ .$$

q.e.d.

This derivation only applies in a point, but since it is frame independent, it implies that test particles follow geodesics. In a different frame, Christoffel symbols will not be zero and we can interpret them as a gravitational potential $\Gamma^{\alpha}_{\mu\nu} \sim g_{\mu\nu,\beta}$.

We can generalise this result also for massless particles, like light. In fact, given the light-cone property $g(\underline{u}, \underline{u})$, which is conserved along a geodesic

$$\nabla_{\underline{u}} g(\underline{u}, \underline{u}) = 0 \ ,$$

we can prove that it implies that it also follows geodesic equation

$$u^{\mu} \nabla_{\mu} u^{\alpha} = 0 \ .$$

Proof. In fact

$$\begin{aligned} 0 &= \nabla_{\underline{u}} g(\underline{u}, \underline{u}) = \nabla_{u^{\mu}\partial_{\mu}} g(\underline{u}, \underline{u}) = u^{\mu} \nabla_{\mu} g(\underline{u}, \underline{u}) \\ &= u^{\mu} g(\nabla_{\mu} \underline{u}, \underline{u}) + u^{\mu} \nabla_{\mu} g(\underline{u}, \nabla_{\mu} \underline{u}) = 2u^{\mu} g(\nabla_{\mu} \underline{u}, \underline{u}) = 2(u^{\mu} \nabla_{\mu} u^{\alpha}) u_{\alpha} \ . \end{aligned}$$

q.e.d.

Therefore, light follows geodesic, even though we cannot identify the proper time as an affine parameter.

2.2 Sources

The metric has 10 independent components, since it is a symmetric 4×4 matrix. We need 10 ten partial differential partial equations at second order. The Einstein tensor could be a valid candidate

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \ ,$$

because it is constructed by contractions of the Riemann tensor. Furthermore, it satisfies the Bianchi identity

$$\nabla_{\mu} G^{\mu\nu} = \nabla_{\nu} G^{\mu\nu} = 0 \ .$$

This implies that the actually degrees of freedom is 6.

The source part of the equation must satisfy the same properties: symmetric and Bianchi identity. By the principle of equivalence, gravitational interactions depends on proper mass, and by special relativity, mass is equivalent to energy. Therefore, the source of gravity is energy. The energy-momentum tensor is a valid candidate. For a perfect fluid, which is a fluid that depends only on proper energy density ρ and proper pressure p, with co-moving 4-velocity u^{μ}, it becomes

$$T^{\mu\nu} = (p + \rho) u^{\mu} u^{\nu} + p g^{\mu\nu} \ .$$

In matrix notation

$$T^{\mu\nu} = \begin{bmatrix} \rho & 0 \\ 0 & pg^{ij} \end{bmatrix} \ , \quad T^{\mu}{}_{\nu} = \begin{bmatrix} -\rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{bmatrix}$$

Notice that pressure is orthogonal to u^μ.

Proof. In fact,

$$(u^\mu u^\nu + g^{\mu\nu})u_\mu = u^\nu \underbrace{u^\mu u_\mu}_{-1} + u^\nu = -u^\nu + u^\nu = 0 \ .$$

q.e.d.

The 4-current $j^\mu = (\rho, p^i)$ satisfies the continuity equation

$$\partial_\mu j^\mu = 0 \ , \quad \partial_t \rho + \underline{\nabla} \cdot \underline{p} = 0 \ ,$$

which physically means that energy is conserved.

The Bianchi identity of the energy-momentum tensor can be decomposed into the conservation of energy and the Euler equation for fluids.

Proof. In fact,

$$\begin{aligned} 0 = \partial_\mu T^{\mu\nu} &= \partial_\mu((p+\rho)u^\mu u^\nu + pg^{\mu\nu}) \\ &= \partial_\mu(p+\rho)u^\mu u^\nu + (p+\rho)(\partial_\mu u^\mu)u^\nu + (p+\rho)u^\mu(\partial_\mu u^\nu) + (\partial_\mu p)g^{\mu\nu} \\ &= u^\mu u^\nu \partial_\mu(p+\rho) + (p+\rho)(u^\nu \partial_\mu u^\mu + u^\mu \partial_\mu u^\nu) + \partial^\nu p \ . \end{aligned}$$

Considering $u^\mu = (1, u^i)$, for the temporal index, we have

$$\begin{aligned} \partial_\mu T^{\mu 0} &= u^\mu \underbrace{u^0}_{1} \partial_\mu(p+\rho) + (p+\rho)(\underbrace{u^0}_{1} \partial_\mu u^\mu + u^\mu \partial_\mu \underbrace{u^0}_{1}) + \partial^0 p \\ &= u^\mu \partial_\mu(p+\rho) + (p+\rho)\partial_\mu u^\mu + \partial^0 p = \partial_\mu((p+\rho)u^\mu) - \dot{p} \\ &= \partial_0(p+\rho) + \partial_i((p+\rho)u^i) - \dot{p} \ , \end{aligned}$$

which in the non-relativistic limit $|p| \ll \rho$ becomes

$$\partial_0(p+\rho) + \partial_i((p+\rho)u^i) - \dot{p} \simeq \partial_0 \rho + \partial_i(\rho u^i) = 0 \ .$$

This is the conservation of energy for a fluid. For the spatial indices, we have

$$\begin{aligned} \partial_\mu T^{\mu i} &= u^\mu u^i \partial_\mu(p+\rho) + (p+\rho)(u^i \partial_\mu u^\mu + u^\mu \partial_\mu u^i) + \partial^i p \\ &= \underbrace{u^0}_{1} u^i \partial_0(p+\rho) + u^j u^i \partial_j(p+\rho) + (p+\rho)(u^i \partial_0 \underbrace{u^0}_{1} + \underbrace{u^0}_{1} \partial_0 u^i) \\ &\quad + (p+\rho)(u^i \partial_j u^j + u^j \partial_j u^i) + \partial^i p \\ &= u^i \partial_0(p+\rho) + u^j u^i \partial_j(p+\rho) + (p+\rho)\partial_0 u^i + (p+\rho)(u^i \partial_j u^j + u^j \partial_j u^i) + \partial^i p \\ &= u^i \underbrace{(\partial_0 \rho + \partial_j(\rho u^j))}_{0} + u^i(\partial_0 p + \partial_j(p u^j)) + (p+\rho)\partial_0 u^i + (p+\rho)(u^i \partial_j u^j + u^j \partial_j u^i) + \partial^i p \\ &= u^i(\partial_0 p + \partial_j(p u^j)) + (p+\rho)\partial_0 u^i + (p+\rho)(u^i \partial_j u^j + u^j \partial_j u^i) + \partial^i p \ , \end{aligned}$$

which in the non-relativistic limit $|p| \ll \rho$ and $u^i \ll c$ becomes

$$\rho\partial_0 u^i + \rho u^j \partial_j u^i + \partial^i p = 0 \ ,$$

This is the Euler equation for a fluid. q.e.d.

Putting together, we have

$$G_{\mu\nu} \propto T_{\mu\nu} \ .$$

The proportionality constant can be found using dimensional analysis and the Newtonian limit. Finally, we find

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ .$$

2.3 Gravity and geometry

The non-linear Einstein field equations tell us that matter curves spacetime, but it follows also geodesic. This means that gravity can be thought as geometry. For example, it can be proved that for dust, i.e. $p = 0$, that the geodesic equation follows from the continuity equation.

Proof. In fact, the energy-momentum tensor is

$$T^{\mu\nu} = \rho u^\mu u^\nu$$

and the continuity equation becomes

$$0 = \nabla_\mu T^{\mu\nu} = \nabla_\mu(\rho u^\mu u^\nu) = \nabla_\mu(\rho u^\mu u^\nu) = (\nabla_\mu \rho) u^\mu u^\nu + \rho(\nabla_\mu u^\mu) u^\nu + (\rho \nabla_\mu u^\nu) u^\mu \ .$$

Now, we multiply for u_ν and we obtain

$$\begin{aligned} 0 = (\nabla_\mu T^{\mu\nu}) u_\nu &= (\nabla_\mu \rho) u^\mu \underbrace{u^\nu u_\nu}_{-1} + \rho(\nabla_\mu u^\mu) \underbrace{u^\nu u_\nu}_{-1} + \rho(\nabla_\mu u^\nu) u^\mu u_\nu \\ &= -(\nabla_\mu \rho) u^\mu - \rho(\nabla_\mu u^\mu) + \rho \underbrace{(\nabla_\mu u^\nu) u_\nu}_{0} u^\mu = -\nabla_\mu(\rho u^\mu) \ , \end{aligned}$$

where we have used the fact that

$$0 = \rho \nabla_\mu (u_\nu u^\nu) = 2\rho(\nabla_\mu u^\nu) u_\nu \ .$$

We invert the first equation

$$\rho(\nabla_\mu u^\nu) u^\mu = -(\nabla_\mu \rho) u^\mu u^\nu - \rho(\nabla_\mu u^\mu) u^\nu = -\nabla_\mu(\rho u^\mu) u^\nu = 0 \ ,$$

which implies that

$$\rho u^\mu \nabla_\mu u^\nu = 0 \ .$$

q.e.d.

Chapter 3

Einstein-Hilbert action

In this chapter, we will study how to derive the Einstein field equations by an action principle and the gauge symmetries of this action by means of diffeomorphisms.

3.1 Einstein-Hilbert action

In the vacuum, Einstein field equations can be obtained by an action, called the Einstein-Hilbert action

$$S = -\int d^4x \ \sqrt{-g}R \ .$$

Proof. We make a variation of the metric $\delta g_{\mu\nu}$, hence

$$\delta(\sqrt{-g}R) = \delta\sqrt{-g}R + \sqrt{-g}\delta R \ .$$

For the first term, we use the identity

$$\delta \det g = \delta \exp \ln \det g = \exp \ln \det g \ \delta \ln \det g = \det g \ \delta \operatorname{tr} \ln g = \det g \ \delta \operatorname{tr}(g^{-1}\delta g) \ ,$$

therefore

$$\begin{aligned} \delta\sqrt{-g} &= \frac{1}{2\sqrt{-g}}\delta\sqrt{-g} = -\frac{1}{2\sqrt{-g}}(-g) \ g^{\mu\nu}\delta g_{\mu\nu} = \frac{1}{2}\sqrt{-g} \ g^{\mu\nu}\delta g_{\mu\nu} \\ &= \frac{1}{2}\sqrt{-g} \ g^{\mu\nu}\delta g_{\mu\nu} = -\frac{1}{2}\sqrt{-g} \ g_{\mu\nu}\delta g^{\mu\nu} \ . \end{aligned} \tag{3.1}$$

For the second term,

$$\delta R = \delta(g^{\mu\nu}R_{\mu\nu}) = \delta g^{\mu\nu}R_{\mu\nu} + g^{\mu\nu}\delta R_{\mu\nu}$$

Now, we vary the Christoffel symbols

$$\delta\Gamma^{\alpha}_{\mu\nu} = \frac{1}{2}\delta(g^{\alpha\beta}(-g_{\mu\nu,\alpha} + g_{\mu\alpha,\nu} + g_{\nu\alpha,\mu})) = \frac{1}{2}g^{\alpha\beta}(-\delta g_{\mu\nu,\alpha} + \delta g_{\mu\alpha,\nu} + \delta g_{\nu\alpha,\mu}) \ . \tag{3.2}$$

Hence,

$$\delta R_{\mu\nu} = \delta\Gamma^{\alpha}_{\mu\nu;\alpha} - \delta\Gamma^{\alpha}_{\mu\alpha;\nu} \ ,$$

and

$$\begin{aligned} g^{\mu\nu}\delta R_{\mu\nu} &= g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu;\alpha} - \delta\Gamma^{\alpha}_{\mu\alpha;\nu} = \nabla_\alpha(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu}) - \nabla_\nu(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\alpha}) \\ &= \nabla_\alpha(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu}) - \nabla_\alpha(g^{\mu\alpha}\delta\Gamma^{\nu}_{\mu\nu}) = \nabla_\alpha(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu} - g^{\mu\alpha}\delta\Gamma^{\nu}_{\mu\nu}) \ , \end{aligned}$$

where we have exchanged α and ν. This term can be neglected because it is a total derivative. Putting together, we obtain

$$\delta(\sqrt{-g}R) = \sqrt{-g}\Big(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\Big)\delta g^{\mu\nu} = \sqrt{-g}G_{\mu\nu}\delta g^{\mu\nu}$$

and

$$\delta S = -\int d^4x \ \sqrt{-g} \ G_{\mu\nu}\delta g^{\mu\nu} \ .$$

Since $\delta g^{\mu\nu}$ is arbitrary, the equations of motion are

$$G_{\mu\nu} = 0 \ .$$

q.e.d.

Considering matter and dimensional analysis as well, the action becomes

$$S = \int d^4x \ \sqrt{-g}\Big(-\frac{R}{16\pi G_N} + \mathcal{L}_M\Big) = S_{EH} + S_M \ ,$$

where $\mathcal{L}_M$ is the Lagrangian density of matter.

Proof. We define the energy-momentum tensor

$$\begin{aligned} T_{\mu\nu} &= \frac{2}{\sqrt{-g}}\frac{\delta S_M}{\delta g^{\mu\nu}} = \frac{2}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}(\sqrt{-g}\mathcal{L}_M) \\ &= \frac{2}{\sqrt{-g}}\sqrt{-g}\frac{\delta\mathcal{L}_M}{\delta g^{\mu\nu}} + \frac{2}{\sqrt{-g}}\mathcal{L}_M\frac{\delta\sqrt{-g}}{\delta g^{\mu\nu}} = 2\frac{\delta\mathcal{L}_M}{\delta g^{\mu\nu}} - \mathcal{L}_M g_{\mu\nu} \ , \end{aligned}$$

where we have used (3.1). Moreover, the variation of the action is

$$\delta S = \int d^4x \ \sqrt{-g}\Big(-\frac{G_{\mu\nu}}{8\pi G_N} + T_{\mu\nu})\delta g^{\mu\nu} \ ,$$

and, since $\delta g^{\mu\nu}$ is arbitrary, the equations of motion are

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ .$$

q.e.d.

3.2 Diffeomorphisms

Consider two charts $x = x^\mu(P)$ and $y = y^\mu(P)$, defined as

$$x\colon \mathcal{D}_x \to \mathbb{R}^n \ , \quad y\colon \mathcal{D}_y \to \mathbb{R}^n \ ,$$

such that they do not overlap $\mathcal{D}_x \cap \mathcal{D}_y \neq \emptyset$. A diffeomorphism is a change of coordinates that can be smoothly connected with the identity

$$y^\mu(P) = x^\mu(P) + \epsilon\xi^\mu(P) \ ,$$

where ϵ is an infinitesimal parameter and ξ^μ are components of a vector field in coordinate basis $\partial/\partial x$

$$\underline{\xi} = \xi^\mu \partial/\partial x^\mu \ .$$

The coordinate basis change as

$$\frac{\partial}{\partial y^\mu} = \frac{\partial x^\nu}{\partial y^\mu}\frac{\partial}{\partial x^\nu} = \left(\delta^\nu{}_\mu + \epsilon\frac{\partial \xi^\nu}{\partial x^\mu}\right)^{-1}\frac{\partial}{\partial x^\nu} = \left(\delta^\nu{}_\mu - \epsilon\frac{\partial \xi^\nu}{\partial x^\mu}\right)\frac{\partial}{\partial x^\nu} \ , \tag{3.3}$$

whereas the dual basis as

$$d\tilde{y}^\mu = \frac{\partial y^\mu}{\partial x^\nu} d\tilde{x}^\nu = \left(\delta^\mu{}_\nu + \epsilon\frac{\partial \xi^\mu}{\partial x^\nu}\right) d\tilde{x}^\nu \ . \tag{3.4}$$

The variation of a scalar $\phi = f \circ x^{-1}$ is

$$\delta\phi(x^\mu) = -\xi^\mu \frac{\partial \phi}{\partial x^\mu} = -\pounds_{\underline{\xi}}\phi \ .$$

This means that shifting the coordinates in the direction of $\underline{\xi}$ is equivalent to dragging backwards the scalar field f.

Proof. Consider $\phi = f \circ x^{-1}$ and $\psi = f \circ y^{-1}$. Since f is a scalar, we have

$$\psi(y^\mu) = \phi(x^\mu) \ .$$

Using the expansions

$$\psi(x^\mu) = \phi(x^\mu) + \epsilon\delta\phi(x^\mu) \ ,$$

$$\psi(y^\mu) = \psi(x^\mu + \epsilon\xi^\mu) \simeq \psi(x^\mu) + \epsilon\xi^\mu\frac{\partial\psi}{\partial x^\mu} \simeq \phi(x^\mu) + \epsilon\delta\phi(x^\mu) + \epsilon\xi^\mu\frac{\partial\phi}{\partial x^\mu} \ ,$$

we have

$$\delta\phi(x^\mu) = -\xi^\mu\frac{\partial\phi}{\partial x^\mu} \ .$$

q.e.d.

The variation of a vector $\underline{v}$ is

$$\delta v^\mu = v^\nu \frac{\partial \xi^\mu}{\partial x^\nu} - \xi^\nu \frac{\partial v^\mu}{\partial x^\nu} = -(\pounds_{\underline{\xi}} \underline{v})^\mu \ .$$

Proof. Since $\underline{v}$ is a scalar, we have

$$V^\nu(y^\mu) \frac{\partial}{\partial y^\nu} = v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} \ .$$

Using (3.3) and the expansions

$$V^\nu(x^\mu) = v^\nu(x^\mu) + \epsilon \delta v^\nu(x^\mu) \ ,$$

$$\begin{aligned}
V^\nu(y^\mu) \frac{\partial}{\partial y^\nu} &= V^\nu(x^\mu + \epsilon \xi^\mu) \frac{\partial}{\partial y^\nu} \simeq \left(V^\nu(x^\mu) + \epsilon \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \right) \left(\delta^\alpha{}_\nu - \epsilon \frac{\partial \xi^\alpha}{\partial x^\nu} \right) \frac{\partial}{\partial x^\alpha} \\
&= \left(v^\nu(x^\mu) + \epsilon \delta v^\nu(x^\mu) + \epsilon \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \right) \left(\delta^\alpha{}_\nu - \epsilon \frac{\partial \xi^\alpha}{\partial x^\nu} \right) \frac{\partial}{\partial x^\alpha} \\
&= v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} + \epsilon \delta v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} + \epsilon \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \frac{\partial}{\partial x^\nu} - \epsilon v^\nu \frac{\partial \xi^\alpha}{\partial x^\nu} \frac{\partial}{\partial x^\alpha} \\
&= v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} + \epsilon \delta v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} + \epsilon \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \frac{\partial}{\partial x^\nu} - \epsilon v^\alpha \frac{\partial \xi^\nu}{\partial x^\alpha} \frac{\partial}{\partial x^\nu} \ ,
\end{aligned}$$

where we have exchanged α and ν, we have

$$\delta v^\nu(x^\mu) = v^\alpha \frac{\partial \xi^\nu}{\partial x^\alpha} - \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \ .$$

q.e.d.

The variation of the metric is

$$\delta v^\mu = v^\nu \frac{\partial \xi^\mu}{\partial x^\nu} - \xi^\nu \frac{\partial v^\mu}{\partial x^\nu} = -(\pounds_{\underline{\xi}} \underline{v})^\mu \ . \tag{3.5}$$

Proof. Since g is a scalar, we have

$$G_{\mu\nu}(y^\mu) d\tilde{y}^\mu d\tilde{y}^\nu = g_{\mu\nu}(x^\mu) d\tilde{x}^\mu d\tilde{x}^\nu \ .$$

Using (3.4) and the expansions

$$G_{\mu\nu}(x^\mu) = g_{\mu\nu}(x^\mu) + \epsilon \delta g_{\mu\nu}(x^\mu) \ ,$$

$$\begin{aligned}
G_{\mu\nu}(y^\mu) d\tilde{y}^\mu d\tilde{y}^\nu &= G_{\mu\nu}(x^\mu + \epsilon \xi^\mu) d\tilde{y}^\mu d\tilde{y}^\nu \\
&= (G_{\mu\nu}(x^\mu) + \epsilon \xi^\alpha g_{\mu\nu,\alpha})(\delta^\mu{}_\alpha + \epsilon \xi^\mu{}_{,\alpha})(\delta^\nu{}_\beta + \epsilon \xi^\nu{}_{,\beta}) d\tilde{x}^\alpha d\tilde{x}^\beta \\
&= (g_{\mu\nu}(x^\mu) + \epsilon \delta g_{\mu\nu}(x^\mu) + \epsilon \xi^\alpha g_{\mu\nu,\alpha})(\delta^\mu{}_\alpha + \epsilon \xi^\mu{}_{,\alpha})(\delta^\nu{}_\beta + \epsilon \xi^\nu{}_{,\beta}) d\tilde{x}^\alpha d\tilde{x}^\beta \\
&= g_{\alpha\beta} d\tilde{x}^\alpha d\tilde{x}^\beta + \epsilon g_{\alpha\nu} \xi^\nu{}_{,\beta} d\tilde{x}^\alpha d\tilde{x}^\beta + \epsilon g_{\mu\beta} \xi^\mu{}_{,\alpha} d\tilde{x}^\alpha d\tilde{x}^\beta \\
&\quad + \epsilon \delta g_{\alpha\beta}(x^\mu) d\tilde{x}^\alpha d\tilde{x}^\beta + \epsilon \xi^\sigma g_{\alpha\beta,\sigma} d\tilde{x}^\alpha d\tilde{x}^\beta \ ,
\end{aligned}$$

we have

$$\begin{aligned}
\delta g_{\alpha\beta}(x^\mu) &= -g_{\alpha\nu}\xi^\nu{}_{,\beta} - g_{\mu\beta}\xi^\mu{}_{,\alpha} - \xi^\sigma g_{\alpha\beta,\sigma} \\
&= -\partial_\beta(g_{\alpha\nu}\xi^\nu) - \partial_\alpha(g_{\mu\beta}\xi^\mu) + g_{\alpha\nu,\beta}\xi^\nu + g_{\mu\beta,\alpha}\xi^\mu - \xi^\sigma g_{\alpha\beta,\sigma} \\
&= -\xi_{\alpha,\beta} - \xi_{\beta,\alpha} + g_{\alpha\nu,\beta}\xi^\nu + g_{\mu\beta,\alpha}\xi^\mu - \xi^\sigma g_{\alpha\beta,\sigma} \\
&= -\xi_{\alpha,\beta} - \xi_{\beta,\alpha} + \xi^\sigma(-g_{\alpha\beta,\sigma} + g_{\alpha\sigma,\beta} + g_{\beta\sigma,\alpha}) \\
&= -\xi_{\alpha,\beta} - \xi_{\beta,\alpha} + \xi_\rho \underbrace{g^{\rho\sigma}(-g_{\alpha\beta,\sigma} + g_{\alpha\sigma,\beta} + g_{\beta\sigma,\alpha})}_{2\Gamma^\rho_{\alpha\beta}} \\
&= -\xi_{\alpha,\beta} - \xi_{\beta,\alpha} + 2\xi_\rho\Gamma^\rho_{\alpha\beta} = -\xi_{\alpha,\beta} + \xi_\rho\Gamma^\rho_{\alpha\beta} - \xi_{\beta,\alpha} + \xi_\rho\Gamma^\rho_{\beta\alpha} = -\xi_{\alpha;\beta} - \xi_{\beta;\alpha} \ ,
\end{aligned}$$

where we have integrated by parts and used the symmetric properties. q.e.d.

3.3 Gauge symmetries

The Einstein-Hilbert action is invariant under these diffeomorphisms. The vacuum action vanishes identically due to the Bianchi identity

$$S_{EH} = \int d^4x \ \sqrt{-g}R \ .$$

Proof. In fact, using (3.5)

$$\begin{aligned}
\delta S_{EH} &= \int d^4x \ \sqrt{-g}G^{\mu\nu}\delta g_{\mu\nu} = -\int d^4x \ \sqrt{-g}G^{\mu\nu}(\nabla_\nu\xi_\mu + \nabla_\mu\xi_\nu) \\
&= \int d^4x \ \sqrt{-g}(\xi_\mu \underbrace{\nabla_\nu G^{\mu\nu}}_{0} + \xi_\nu \underbrace{\nabla_\mu G^{\mu\nu}}_{0}) = 0 \ ,
\end{aligned}$$

where we have integrated by parts. q.e.d.

The matter action vanishes identically due to the continuity equation

$$S_{EH} = \int d^4x \ \sqrt{-g}\mathcal{L}_M \ .$$

Proof. In fact, using (3.5)

$$\begin{aligned}
\delta S_{EH} &= \int d^4x \ \sqrt{-g}T^{\mu\nu}\delta g_{\mu\nu} = -\int d^4x \ \sqrt{-g}T^{\mu\nu}(\nabla_\nu\xi_\mu + \nabla_\mu\xi_\nu) \\
&= \int d^4x \ \sqrt{-g}(\xi_\mu \underbrace{\nabla_\nu T^{\mu\nu}}_{0} + \xi_\nu \underbrace{\nabla_\mu T^{\mu\nu}}_{0}) = 0 \ ,
\end{aligned}$$

where we have integrated by parts. q.e.d.

Furthermore, the Riemann tensor is invariant under diffeomorphisms.

Proof. In fact, using (3.2) and (3.5),

$$
\begin{aligned}
\delta R^{\mu}{}_{\nu\alpha\beta} &= -\delta\Gamma^{\mu}_{\nu\alpha,\beta} + \delta\Gamma^{\mu}_{\nu\beta,\alpha} \\
&= -\frac{1}{2}g^{\mu\rho}\Big(- \delta g_{\nu\alpha,\rho\beta} + \delta g_{\nu\rho,\alpha\beta} + \delta g_{\alpha\rho,\nu\beta}\Big) + \frac{1}{2}g^{\mu\rho}\Big(- \delta g_{\nu\beta,\rho\alpha} + \delta g_{\nu\rho,\beta\alpha} + \delta g_{\beta\rho,\nu\alpha}\Big) \\
&= \frac{1}{2}g^{\mu\rho}\Big(\delta g_{\nu\alpha,\rho\beta} - \delta g_{\nu\rho,\alpha\beta} - \delta g_{\alpha\rho,\nu\beta} - \delta g_{\nu\beta,\rho\alpha} + \delta g_{\nu\rho,\beta\alpha} + \delta g_{\beta\rho,\nu\alpha}\Big) \\
&= -\frac{1}{2}g^{\mu\rho}\Big(\cancel{\xi_{\nu,\alpha\rho\beta}} + \cancel{\xi_{\alpha,\nu\rho\beta}} - \cancel{\xi_{\nu,\rho\alpha\beta}} - \cancel{\xi_{\rho,\nu\alpha\beta}} - \cancel{\xi_{\alpha,\rho\nu\beta}} - \cancel{\xi_{\rho,\alpha\beta\nu}} \\
&\qquad - \cancel{\xi_{\nu,\beta\rho\alpha}} - \cancel{\xi_{\beta,\nu\rho\alpha}} + \cancel{\xi_{\nu,\rho\beta\alpha}} + \cancel{\xi_{\rho,\nu\beta\alpha}} + \cancel{\xi_{\beta,\rho\nu\alpha}} + \cancel{\xi_{\rho,\beta\nu\alpha}}\Big) = 0 \ ,
\end{aligned}
$$

where we have used the fact that partial derivatives commute. q.e.d.

Part II

Gravitational waves

Chapter 4

Linearised Einstein field equations

In this chapter, we will study the weak-field limit, in order to find the linearised Einstein field equations in the De Donder gauge. With the addition of non-relativistic limit, we will recover the experimentally tested Newtonian laws (gravity and second law of dynamics).

4.1 Linearised Einstein field equations

Consider the weak field limit, in which local curvature is small. This means that there exists a reference frame such that

$$g_{\mu\nu} = \eta_{\mu\nu} + \epsilon h_{\mu\nu} \ ,$$

where ϵ is an infinitesimal parameter. The linearised Einstein field equations in first order in ϵ are

$$\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} = 16\pi G_N \epsilon T_{\mu\nu} \ .$$

Proof. The Christoffel symbols in first order in ϵ are

$$\begin{aligned}
\Gamma^\alpha_{\mu\nu} &= \frac{1}{2}g^{\alpha\beta}(g_{\mu\beta,\nu} + g_{\nu\beta,\mu} - g_{\mu\nu,\beta}) \\
&= \frac{1}{2}g^{\alpha\beta}(\cancel{\eta_{\mu\beta,\nu}} + \epsilon h_{\mu\beta,\nu} + \cancel{\eta_{\nu\beta,\mu}} + \epsilon h_{\nu\beta,\mu} - \cancel{\eta_{\mu\nu,\beta}} - \epsilon h_{\mu\nu,\beta}) \\
&= \frac{\epsilon}{2}g^{\alpha\beta}(h_{\mu\beta,\nu} + h_{\nu\beta,\mu} - h_{\mu\nu,\beta}) \\
&\simeq \frac{\epsilon}{2}(\eta^{\alpha\beta} - \epsilon h^{\alpha\beta})(h_{\mu\beta,\nu} + h_{\nu\beta,\mu} - h_{\mu\nu,\beta}) \\
&\simeq \frac{1}{2}(\epsilon\eta^{\alpha\beta} - \cancel{\epsilon^2 h^{\alpha\beta}})(h_{\mu\beta,\nu} + h_{\nu\beta,\mu} - h_{\mu\nu,\beta}) \\
&\simeq \frac{\epsilon}{2}\eta^{\alpha\beta}(h_{\mu\beta,\nu} + h_{\nu\beta,\mu} - h_{\mu\nu,\beta}) \ ,
\end{aligned} \tag{4.1}$$

where we have Taylor expanded the inverse of the metric

$$g^{\alpha\beta} = (g_{\alpha\beta})^{-1} = (\eta_{\alpha\beta} + \epsilon h_{\alpha\beta})^{-1} \simeq \eta_{\alpha\beta} - \epsilon h_{\alpha\beta} \ .$$

The Riemann tensor in first order in ϵ is

$$\begin{aligned}
R^{\mu}{}_{\nu\alpha\beta} &= \Gamma^{\mu}_{\nu\beta,\alpha} - \Gamma^{\mu}_{\nu\alpha,\beta} + \Gamma^{\lambda}_{\nu\beta}\Gamma^{\mu}_{\lambda\alpha} - \Gamma^{\lambda}_{\nu\alpha}\Gamma^{\mu}_{\lambda\beta} \\
&\simeq \Gamma^{\mu}_{\nu\beta,\alpha} - \Gamma^{\mu}_{\nu\alpha,\beta} \\
&= \partial_\alpha(\frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\nu\lambda,\beta} + h_{\beta\lambda,\nu} - h_{\nu\beta,\lambda})) - \partial_\beta(\frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\nu\lambda,\alpha} + h_{\alpha\lambda,\nu} - h_{\nu\alpha,\lambda})) \\
&= \frac{\epsilon}{2}\cancel{\eta^{\mu\lambda}{}_{,\alpha}}(h_{\nu\lambda,\beta} + h_{\beta\lambda,\nu} - h_{\nu\beta,\lambda}) + \frac{\epsilon}{2}\eta^{\mu\lambda}(\cancel{h_{\nu\lambda,\beta\alpha}} + h_{\beta\lambda,\nu\alpha} - h_{\nu\beta,\lambda\alpha}) \\
&\quad - \frac{\epsilon}{2}\cancel{\eta^{\mu\lambda}{}_{,\beta}}(h_{\nu\lambda,\alpha} + h_{\alpha\lambda,\nu} - h_{\nu\alpha,\lambda}) - \frac{\epsilon}{2}\eta^{\mu\lambda}(\cancel{h_{\nu\lambda,\alpha\beta}} + h_{\alpha\lambda,\nu\beta} - h_{\nu\alpha,\lambda\beta}) \\
&= \frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\beta\lambda,\nu\alpha} - h_{\nu\beta,\lambda\alpha}) - \frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\alpha\lambda,\nu\beta} - h_{\nu\alpha,\lambda\beta}) \\
&= \frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\beta\lambda,\nu\alpha} - h_{\nu\beta,\lambda\alpha} - h_{\alpha\lambda,\nu\beta} + h_{\nu\alpha,\lambda\beta}) \ .
\end{aligned}$$

The Ricci tensor in first order in ϵ is

$$\begin{aligned}
R_{\nu\beta} &= R^{\mu}{}_{\nu\mu\beta} \\
&= \frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\beta\lambda,\nu\mu} - h_{\nu\beta,\lambda\mu} - h_{\mu\lambda,\nu\beta} + h_{\nu\mu,\lambda\beta}) \\
&= \frac{\epsilon}{2}\eta^{\mu\lambda}(\partial_\nu\partial_\mu h_{\beta\lambda} - \partial_\lambda\partial_\mu h_{\nu\beta} - \partial_\nu\partial_\beta h_{\mu\lambda} + \partial_\lambda\partial_\beta h_{\nu\mu}) \\
&= \frac{\epsilon}{2}(\partial_\nu\partial^\mu h_{\beta\mu} - \partial^\mu\partial_\mu h_{\nu\beta} - \partial_\nu\partial_\beta h^{\mu}{}_{\mu} + \partial^\mu\partial_\beta h_{\nu\mu}) \\
&= \frac{\epsilon}{2}(\partial^\mu\partial_\beta h_{\nu\mu} + \partial^\mu\partial_\nu h_{\beta\mu} - \partial_\nu\partial_\beta h - \Box h_{\nu\beta}) \ ,
\end{aligned}$$

where we have set $\lambda = \mu$. The Ricci scalar in first order in ϵ is

$$\begin{aligned}
R &= R^{\nu}{}_{\nu} \\
&= \eta^{\nu\beta}R_{\nu\beta} \\
&= \frac{\epsilon}{2}\eta^{\nu\beta}(\partial^\mu\partial_\beta h_{\nu\mu} + \partial^\mu\partial_\nu h_{\beta\mu} - \partial_\nu\partial_\beta h - \Box h_{\nu\beta}) \\
&= \frac{\epsilon}{2}(\partial^\mu\partial^\nu h_{\nu\mu} + \partial^\mu\partial^\beta h_{\beta\mu} - \partial_\nu\partial^\nu h - \Box h^{\nu}{}_{\nu}) \\
&= \frac{\epsilon}{2}(2\partial^\mu\partial^\nu h_{\nu\mu} - 2\Box h) \\
&= \epsilon(\partial^\mu\partial^\nu h_{\nu\mu} - \Box h) \ ,
\end{aligned}$$

where we have set $\beta = \nu$. Finally, the Einstein tensor in first order in ϵ is

$$\begin{aligned}
G_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \\
&= \frac{\epsilon}{2}(\partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu}) - \frac{1}{2}\epsilon(\partial^\alpha\partial^\beta h_{\alpha\beta} - \Box h)g_{\mu\nu} \\
&= \frac{\epsilon}{2}(\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta}) \ .
\end{aligned}$$

The linearised Einstein field equations are

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ ,$$

$$\frac{\epsilon}{2}(\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta}) = 8\pi G_N \epsilon\epsilon T_{\mu\nu} \ ,$$

$$\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} = 16\pi G_N \epsilon T_{\mu\nu} \ ,$$

where we have expanded ${T'}_{\mu\nu} = T^{(0)}_{\mu\nu} + \epsilon T_{\mu\nu}$ and noticed that $T^{(0)}_{\mu\nu} = 0$, because the Minkowski metric solves Einstein equations. q.e.d.

Since the metric is invariant under a gauge symmetry, following by the Bianchi identity, we can choose a suitable gauge fixing to simplify these equations: the De Donder gauge

$$\partial^\mu h_{\mu\nu} = \frac{1}{2}\partial_\nu h \ .$$

Therefore, the linearised Einstein field equations becomes

$$-\Box h_{\mu\nu} = 16\pi G_N (T_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} T) \ .$$

Proof. In fact, the right side

$$\begin{aligned}
G_{\mu\nu} &= \eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} \\
&= \eta_{\mu\nu}\partial^\alpha\partial_\alpha h - \partial^\alpha\partial_\alpha h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} \\
&= \eta_{\mu\nu}\partial^\alpha\partial_\alpha h - \partial^\alpha\partial_\alpha h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha \underbrace{\partial^\beta h_{\beta\alpha}}_{\frac{1}{2}\partial_\alpha h} + \partial_\nu \underbrace{\partial^\beta h_{\beta\mu}}_{\frac{1}{2}\partial_\mu h} + \partial_\mu \underbrace{\partial^\beta h_{\beta\nu}}_{\frac{1}{2}\partial_\nu h} \\
&= \eta_{\mu\nu}\partial^\alpha\partial_\alpha h - \partial^\alpha\partial_\alpha h_{\mu\nu} - \partial_\mu\partial_\nu h - \frac{1}{2}\eta_{\mu\nu}\partial^\alpha\partial_\alpha h + \frac{1}{2}\partial_\mu h + \frac{1}{2}\partial_\mu\partial_\nu h \\
&= \eta_{\mu\nu}\partial^\alpha\partial_\alpha h - \partial^\alpha\partial_\alpha h_{\mu\nu} - \cancel{\partial_\mu\partial_\nu h} - \frac{1}{2}\eta_{\mu\nu}\partial^\alpha\partial_\alpha h + \cancel{\frac{1}{2}\partial_\nu\partial_\mu h} + \cancel{\frac{1}{2}\partial_\mu\partial_\nu h} \\
&= -\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}\Box h \ .
\end{aligned}$$

Hence,

$$\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}\Box h = 16\pi G_N T_{\mu\nu} \ .$$

Furthermore, the trace of the Einstein tensor is

$$\begin{aligned}
\eta^{\mu\nu}G_{\mu\nu} &= \eta^{\mu\nu}(\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta}) \\
&= \underbrace{\eta^{\mu\nu}\eta_{\mu\nu}}_{4}\Box h - \Box\underbrace{\eta^{\mu\nu}h_{\mu\nu}}_{h} - \eta^{\mu\nu}\partial_\mu\partial_\nu h - \underbrace{\eta^{\mu\nu}\eta_{\mu\nu}}_{4}\partial^\alpha\partial^\beta h_{\alpha\beta} + \eta^{\mu\nu}\partial^\beta\partial_\nu h_{\mu\beta} + \eta^{\mu\nu}\partial^\beta\partial_\mu h_{\nu\beta} \\
&= 4\Box h - \Box h - \partial^\nu\partial_\nu h - 4\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial^\mu h_{\mu\beta} + \partial^\beta\partial^\nu h_{\nu\beta} \\
&= 2\Box h - 2\partial^\nu\underbrace{\partial^\mu h_{\mu\nu}}_{\frac{1}{2}\partial_\nu h} \\
&= \Box h \ ,
\end{aligned}$$

which is equal to

$$16\pi G_N \underbrace{\eta^{\mu\nu}T_{\mu\nu}}_{T} = 16\pi G_N T \ .$$

Putting together

$$\begin{aligned}
-\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}\Box h &= 16\pi G_N T_{\mu\nu} \ , \\
-\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}(16\pi G_N T) &= 16\pi G_N T_{\mu\nu} \ , \\
-\Box h_{\mu\nu} &= 16\pi G_N\Big(T_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}T) \ .
\end{aligned}$$

q.e.d.

4.2 Newtonian regime

In order to recover the full Newtonian regime, we need also to ask that the matter source is static and the only non-vanishing component of the energy momentum tensor is

$$T^{\mu\nu} \simeq T_{00} = -T \ .$$

Therefore, we recover the Poisson equation

$$\nabla^2 V_N = 4\pi G_N \rho$$

where $h_{00} = -2V_N$ and G_N is indeed the Newton constant.

Proof. In fact

$$T^{00}(\underline{x}) = (\rho + p)u^0u^0 + p\eta^{00} = \rho(\underline{x}) \ .$$

Hence,

$$\Box h_{00}(\underline{x}) \simeq \nabla^2 h_{00}(\underline{x}) = -8\pi G_N T_{00}(\underline{x}) = -8\pi G_N \rho(\underline{x}) \ .$$

We identify $h_{00} = -2V_N$ and we obtain

$$\nabla^2 V_N = 4\pi G_N \rho \ .$$

q.e.d.

Furthermore, if we consider the non-relativistic limit, in which particles are not static, but move slower $u^i = \epsilon v^i \ll c$. The geodesic equation becomes

$$\frac{d^2x^i}{dt^2} \simeq -\frac{\partial V_N}{\partial x^i} \ .$$

Proof. We expand the 4-velocity $u^\mu = (1, \epsilon v^i)$ so that

$$\frac{d^2x^\mu}{d\tau^2} = \epsilon\Big(0, \frac{dv^i}{dt}\Big) = \epsilon\Big(0, \frac{d^2x^i}{dt^2}\Big) \ .$$

Using (4.1), we find

$$\begin{aligned}
\Gamma^\beta_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} &= \frac{\epsilon}{2}\eta^{\alpha\beta}(-h_{\mu\nu,\alpha} + h_{\mu\alpha,\nu} + h_{\nu\alpha,\mu})\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} \\
&= \frac{\epsilon}{2}\eta^{\alpha\beta}(-h_{\mu\nu,\alpha} + h_{\mu\alpha,\nu} + h_{\nu\alpha,\mu})\delta^\mu{}_0\delta^\nu{}_0 \\
&= \frac{\epsilon}{2}\eta^{\alpha\beta}(-h_{00,\alpha} + \underbrace{h_{0\alpha,0}}_{0} + \underbrace{h_{0\alpha,0}}_{0}) \\
&= -\frac{\epsilon}{2}\eta^{\alpha\beta}h_{00,\alpha}
\end{aligned}$$

where we have neglected terms of higher order. Putting together

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} \simeq \epsilon\Big(\frac{d^2x^i}{dt^2} + \frac{1}{2}h_{00,i}\Big)$$

and using $h_{00} = -2V_N$, we find

$$\frac{d^2x^i}{dt^2} \simeq -\frac{\partial V_N}{\partial x^i} \ .$$

q.e.d.

Chapter 5

Gravitational waves

In this chapter, we will study how to solve the linearised Einstein field equations with gravitational plane waves: form, production and detection of them.

5.1 Plane waves in vacuum

In is useful to express the perturbation part of the metric tensor h, in terms of the transverse tensor

$$h^T_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \ .$$

It is equivalent, because once we find the transverse tensor, we can always recover the original one. The De Donder gauge becomes the transverse gauge

$$\partial^\mu h^T_{\mu\nu} = 0 \ .$$

It is also called the trace-inverse, since

$$\eta^{\mu\nu}h^T_{\mu\nu} = \eta^{\mu\nu}h_{\mu\nu} - \frac{1}{2}\underbrace{\eta^{\mu\nu}\eta_{\mu\nu}}_{4}h = h - 2h = -h \ .$$

The linearised Einstein field equations becomes

$$-\Box h^T_{\mu\nu} = 16\pi G_N T_{\mu\nu} \ .$$

Proof. In fact

$$16\pi G_N T_{\mu\nu} = -\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}\Box h = -\Box\Big(h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h\Big) = -\Box h^T_{\mu\nu} \ .$$

q.e.d.

Now, we consider the vacuum case. For massless gravitational waves, the equations of motion become

$$\Box h^T_{\mu\nu} = 0 \ .$$

However, we notice that there is still a residual gauge, always inside the De Donder one. In fact, if we make another diffeomorphism transformation

$$\tilde{h}_{\mu\nu} = h_{\mu\nu} + \xi_{\mu,\nu} + \xi_{\nu,\mu} \ ,$$

where ξ^μ is a harmonic, i.e. $\Box\xi^\mu = 0$, we find that both the equations of motion and the transverse gauge are still satisfied.

Proof. For the equations of motion, we have

$$\begin{aligned}
\Box\tilde{h}^T_{\mu\nu} &= \Box\tilde{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\Box\tilde{h} \\
&= \Box h_{\mu\nu} + \underbrace{\Box\xi_{\mu,\nu}}_{0} + \underbrace{\Box\xi_{\nu,\mu}}_{0} - \frac{1}{2}\eta_{\mu\nu}\Box h - \frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\underbrace{\Box\xi_{\alpha,\beta}}_{0} - \frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\underbrace{\Box\xi_{\beta,\alpha}}_{0} \\
&= \Box h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\Box h = \Box h^T_{\mu\nu} = 0 \ .
\end{aligned}$$

For the transverse gauge, we have

$$\begin{aligned}
\partial^\mu\tilde{h}^T_{\mu\nu} &= \partial^\mu\tilde{h}_{\mu\nu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\tilde{h} \\
&= \partial^\mu h_{\mu\nu} + \partial^\mu\xi_{\mu,\nu} + \partial^\mu\xi_{\nu,\mu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}h - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\alpha,\beta} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\beta,\alpha} \\
&= \underbrace{\partial^\mu h_{\mu\nu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}h}_{0} + \partial^\mu\xi_{\mu,\nu} + \partial^\mu\xi_{\nu,\mu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\alpha,\beta} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\beta,\alpha} \\
&= \partial^\mu\xi_{\mu,\nu} + \partial^\mu\xi_{\nu,\mu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\alpha,\beta} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\beta,\alpha} \\
&= \eta^{\alpha\beta}\left(\cancel{\xi_{\beta,\nu\alpha}} + \xi_{\nu,\beta\alpha} - \cancel{\frac{1}{2}\xi_{\alpha,\beta\nu}} - \cancel{\frac{1}{2}\xi_{\beta,\alpha\nu}}\right) = \Box\xi_\nu = 0 \ .
\end{aligned}$$

q.e.d.

Now, consider a plane wave ansatz

$$h^T_{\mu\nu}(x) = \epsilon_{\mu\nu}\exp(ik_\alpha x^\alpha) \ ,$$

where $\epsilon_{\mu\nu}$ are the components of a symmetric polarisation tensor

$$\epsilon_{\mu\nu} =$$

Using the equations of motion, we find the dispersion relation for a massless wave

$$k^\alpha k_\alpha = 0 \ ,$$

which shows that gravitational waves travel at the speed of light.

Proof. In fact,

$$\Box h^T_{\mu\nu} = -k^\alpha k_\alpha h^T_{\mu\nu} = 0 \ .$$

q.e.d.

Using the transverse gauge, we find the transversality condition

$$k^\mu \epsilon_{\mu\nu} = k^\nu \epsilon_{\mu\nu} = 0 \ .$$

Proof. In fact,

$$\partial^\mu h^T_{\mu\nu} = ik^\mu \epsilon_\mu \exp(ik^\alpha x_\alpha) = 0 \ .$$

q.e.d.

Consider a plane wave moving along the z-axis, with wave vector $k^\mu = (k, 0, 0, k)$. This implies that for a generic vector v

$$0 = k_\mu v^\mu = k(v^0 - v^3) \ , \quad v^0 = v^3 \ .$$

Therefore, the components of the polarisation tensor are now

$$\epsilon_{\mu\nu} = \begin{bmatrix} a & b & c & a \\ b & e & f & b \\ c & f & h & c \\ a & b & c & a \end{bmatrix} \ ,$$

where we have used $\epsilon_{0\mu} = \epsilon_{\mu 0} = \epsilon_{3\mu} = \epsilon_{\mu 3}$. However, we still have a residual gauge freedom to satisfies: we choose 3 degrees of freedom to put $a = b = c = 0$ and the remaining one to set the trace equals to zero $e + f = 0$. Hence

$$\epsilon_{\mu\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & e & f & 0 \\ 0 & f & -e & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \ .$$

Putting together, we have

$$h^{TT}_{\mu\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & e & f & 0 \\ 0 & f & -e & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \exp(-ikt + ikz) \ .$$

Now, we can write the general gravitational plane wave as linear combination of

$$\begin{aligned}
h^{TT}_{\mu\nu} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & e & f & 0 \\ 0 & f & -e & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \exp(-ikt + ikz) \\
&= e \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \exp(-ikt + ikz) + f \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \exp(-ikt + ikz) \\
&= e\epsilon_+ \exp(-ikt + ikz) + f\epsilon_\times \exp(-ikt + ikz) \ ,
\end{aligned}$$

where we have defined the

$$\epsilon_+ = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \ , \quad \epsilon_\times = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \ .$$

Now, we take the real part, and we find

$$\begin{aligned}
\mathrm{Re}\, h^{TT}_{\mu\nu} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & e\sin(kt - kz) & 0 & 0 \\ 0 & 0 & -e\sin(kt - kz) & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & f\sin(kt - kz) & 0 \\ 0 & f\sin(kt - kz) & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
&= \mathrm{Re}\, h^{TT}_{\mu\nu\ +} + \mathrm{Re}\, h^{TT}_{\mu\nu\ \times} \ .
\end{aligned}$$

Consider now a free-falling test particle located at $x = 0$ in space, we find that a circle will turn into an ellipse along x and y

$$ds^2_+ = (\eta_{\mu\nu} + \mathrm{Re}\, h^{TT}_{\mu\nu\ +})dx^\mu dx^\nu = -dt^2 + (1 + e\sin(kt))dx^2 + (1 - e\sin(kt))dy^2 + dz^2 \ ,$$

or diagonally

$$ds^2_\times = (\eta_{\mu\nu} + \mathrm{Re}\, h^{TT}_{\mu\nu\ \times})dx^\mu dx^\nu = -dt^2 + dx^2 + dy^2 + 2f\sin(kt)dxdy + dz^2 \ .$$

We can show that trajectories at constant x, y and $z = 0$ are indeed geodesics.

Proof. In fact, consider the Lagrangian for a particle of mass m of the metric g_+

$$L = \frac{m}{2}\Big(- \dot{t}^2 + (1 + a\sin(kt))\dot{x}^2 + (1 - a\sin(kt))\dot{x}^2\Big)\dot{y}^2 + \dot{z}^2 \ .$$

Since L does not depend on x, y and z, their momenta are conserved and they can be set to zero with suitable initial conditions. Furthermore, because $L = -m/2$, we have $\dot{t}^2 = 1$, which solved means $t = \tau + t_0$. We have built a locally inertial reference frame (t, x, y, z). q.e.d.

However, their velocity does not vanish.

Proof. In fact, for a particle located in $x = l_0$ and $y = z = 0$, we have

$$l = \int_0^{l_0} dx g_{xx} = \int_0^{l_0} dx \sqrt{1 + e\sin(kt)} = \sqrt{1 + e\sin(kt)} l_0 \simeq (1 + \frac{e}{2}\sin(kt)) l_0 \ ,$$

where we have used $e \ll 1$ for weak field. Therefore, the velocity is

$$v = \frac{dl}{dt} \simeq \frac{e}{2} k \cos(kt) l_0 \ .$$

q.e.d.

5.2 Production

Consider four particles of mass m on the plane xy force to oscillate as a gravitational wave. In locally inertial frame, we have

$$x_1 = -x_2 = l_0 + \Delta l \sin(\omega t) \ ,$$

$$y_3 = -y_4 = l_0 - \Delta l \sin(\omega t) \ ,$$

$$y_1 = y_2 = x_3 = x_4 = 0 \ ,$$

where $i = 1, 2, 3, 4$. The center of mass is fixed and the total mass of the system is $M = 4m$. The total dipole moment is

$$D = m\Big(\sum_{i=1}^{2} x_i(t) + \sum_{i=3}^{4} y_i(t)\Big) = 0 \ ,$$

whereas the quadrupole moment is

$$Q(t) = m\Big(\sum_{i=1}^{2} x_i^2(t) + \sum_{i=3}^{4} y_i^2(t)\Big) = M(l_0 + \Delta l \sin(\omega t))^2 \ .$$

The total kinetic energy is

$$E = \frac{m}{2}\Big(\sum_{i=1}^{2} \dot{x}_i^2(t) + \sum_{i=3}^{4} \dot{y}_i^2(t)\Big) = \frac{M}{2}\Delta l^2 \omega^2 \cos^2(\omega t) \ .$$

Comparing it with the energy of a gravitational wave, we find

$$E_{gw} = \frac{1}{2}(4m)v^2 = \frac{M}{8} l_0^2 a^2 k^2 \sin^2(kt + t_0) \ ,$$

hence $\omega = k$, $t_0 = \pi/2$ and the strain is $a \sim \delta l / l_0 \sim \Delta l / l$.

5.3 Quadrupole formula

We can make this analysis more formally. Consider the linearised Einstein equation in transverse gauge

$$-\Box h^T_{\mu\nu} = 16\pi G_N T_{\mu\nu} \ .$$

The solution for this equation can be obtained by means of the Green function $G(x-y)$, which satisfies the Green equation

$$\Box_x G(x-y) = \delta^4(x-y) \ ,$$

where x and y are spacetime 4-vectors. The general solution of the Einstein equations is given by

$$h^T_{\mu\nu} = -16\pi G_N \int d^4y \ G(x-y) T_{\mu\nu}(y) \ .$$

where we have ignored the factor $\sqrt{-g}$ since we are linear. Green function depends on the boundary conditions. We choose the retarded Green function, since we want to study the effects of signal after it is propagated in the past. The Green function of the d'Alambertian is given by

$$G(x-y) = -\frac{1}{4\pi|\underline{x}-\underline{y}|}\theta(x^0-y^0)\delta(|\underline{x}-\underline{y}|-(x^0-y^0)) \ .$$

Therefore,

$$\begin{aligned} h^T_{\mu\nu} &= -16\pi G_N \int d^4y \ G(x-y) T_{\mu\nu}(y) \\ &= -16\pi G_N \int d^4y \ T_{\mu\nu}(y)\Big(-\frac{1}{4\pi|\underline{x}-\underline{y}|}\theta(x^0-y^0)\delta(|\underline{x}-\underline{y}|-(x^0-y^0))\Big) \\ &= 4G_N \int d^3y \ \frac{1}{|\underline{x}-\underline{y}|} T_{\mu\nu}(t_r, \underline{y}) \ , \end{aligned}$$

where $t_r = t - |\underline{x}-\underline{y}|$ is the retarded time. Physically, it means that the effect of the gravitational disturbance at a point $t, \underline{x}$ can be traced back to points on the past light cone $(t_r, \underline{x}-\underline{y})$. Now, we take the Fourier transform of the metric perturbation

$$\begin{aligned} \tilde{h}^T_{\mu\nu}(\omega, \underline{x}) &= \int \frac{dt}{(2\pi)^{1/2}} \exp(i\omega t) h^T_{\mu\nu}(t, \underline{x}) \\ &= 4G_N \int \frac{dt}{(2\pi)^{1/2}} d^3y \ \exp(i\omega t) \frac{1}{|\underline{x}-\underline{y}|} T_{\mu\nu}(t_r, \underline{y}) \end{aligned}$$

We make a change of variable into t_r and we find

$$\begin{aligned}\tilde{h}^T_{\mu\nu}(\omega,\underline{x}) &= 4G_N \int \frac{dt_r}{(2\pi)^{1/2}} d^3y \ \exp(i\omega(t_r+|\underline{x}-\underline{y}|))\frac{1}{|\underline{x}-\underline{y}|}T_{\mu\nu}(t_r,\underline{y}) \\ &= 4G_N \underbrace{\int \frac{dt_r}{(2\pi)^{1/2}} \ \exp(i\omega t_r)T_{\mu\nu}(t_r,\underline{y})}_{\tilde{T}_{\mu\nu}(\omega,\underline{y})} \int d^3y \exp(i\omega|\underline{x}-\underline{y}|)\frac{1}{|\underline{x}-\underline{y}|} \\ &= 4G_N \int d^3y \ \exp(i\omega|\underline{x}-\underline{y}|)\frac{1}{|\underline{x}-\underline{y}|}\tilde{T}_{\mu\nu}(\omega,\underline{y}) \ .\end{aligned}$$

So far, we have made exact computations. Now, we consider gravitational waves emitted by a non-relativistic isolated far-away source. We individuate the size of the source to be $r+\delta r$ which low frequency of emission $\delta r \ll 1/\omega$. Physically, it means that light travels faster than the components of the source. This means that

$$\frac{\exp(i\omega|\underline{x}-\underline{y}|)}{|\underline{x}-\underline{y}|} \simeq \frac{\exp(i\omega r)}{r} \ ,$$

hence

$$\tilde{h}^T_{\mu\nu}(\omega,\underline{x}) \simeq 4G_N \frac{\exp(i\omega r)}{r} \int d^3y \ \tilde{T}_{\mu\nu}(\omega,\underline{y}) \ .$$

Using the transverse gauge, we need only to find the spatial components. In fact, we have

$$\partial^\mu \tilde{h}_{\mu\nu}(t,\mathbf{x}) = 0 \ , \quad \tilde{h}^T_{0\nu} = -\frac{i}{\omega}\partial^i \tilde{h}_{i\nu} \ .$$

Let us restrict ourselves to the spatial components. Notice that

$$0 = \int d^3y \ \partial_k(y^i\tilde{T}^{kj}) = \int d^3y \ y^i\partial_k(\tilde{T}^{kj}) + \int d^3y \ \tilde{T}^{ij} \ ,$$

because it is a boundary term. Therefore, we have

$$\int d^3y \ \tilde{T}^{ij} = -\int d^3y \ y^i\partial_k(\tilde{T}^{kj}) \ . \tag{5.1}$$

Using the continuity equation, we find

$$\partial^\mu \tilde{T}_{\mu\nu} = 0 \ , \quad -\partial_k \tilde{T}^{k\nu} = i\omega T^{0\nu} \ .$$

Therefore,

$$\begin{aligned}\int d^3y \ T^{ij}(\omega,\underline{y}) &= -\int d^3y \ y^i\partial_k(\tilde{T}^{kj}) = i\omega \int d^3y \ y^i\tilde{T}^{0j} \\ &= \frac{i\omega}{2}\int d^3y \ (y^i\tilde{T}^{0j}+y^j\tilde{T}^{0i})\end{aligned}$$

where we have symmetrised, since T is symmetric. Exploiting again (5.1), we find

$$\int d^3y\ T^{ij}(\omega,\underline{y}) = \frac{i\omega}{2}\int d^3y\ (\partial_l(y^iy^l\tilde{T}^{0j}) - y^iy^j\partial_l\tilde{T}^{0l}) = -\frac{\omega^2}{2}\int d^3y\ y^iy^j\tilde{T}^{00}\ .$$

We define the quadrupole moment tensor of the energy density of the source

$$I^{ij}(t) = \int d^3y\ y^iy^jT^{00}\ .$$

Putting together, we obtain

$$\int d^3y\ T^{ij}(\omega,\underline{y}) = -\frac{\omega^2}{2}\int d^3y\ y^iy^j\tilde{T}^{00} = -\frac{\omega^2}{2}I^{ij}\ ,$$

$$\begin{aligned}\tilde{h}^T_{ij}(\omega,\underline{x}) &\simeq 4G_N\frac{\exp(i\omega r)}{r}\int d^3y\ \tilde{T}_{ij}(\omega,\underline{y})\\ &= -\frac{\omega^2}{2}\frac{\exp(i\omega r)}{r}4G_NI^{ij} = -2G_N\omega^2\frac{\exp(i\omega r)}{r}\tilde{I}_{ij}(\omega)\ .\end{aligned}$$

Finally, we find the quadrupole formula by an inverse Fourier transform

$$\begin{aligned}h^T_{ij}(t,\underline{x}) &= \int\frac{d\omega}{(2\pi)^{1/2}}\exp(-i\omega t)\tilde{h}^T_{ij}(\omega,\underline{x})\\ &= -2G_N\int\frac{d\omega}{(2\pi)^{1/2}}\exp(-i\omega t)\omega^2\frac{\exp(i\omega r)}{r}\tilde{I}_{ij}(\omega)\\ &= -\frac{2G_N}{r}\int\frac{d\omega}{(2\pi)^{1/2}}\omega^2\exp(-i\omega(t-r))\tilde{I}_{ij}\\ &= -\frac{2G_N}{r}\frac{d^2}{dt^2}I_{ij}(t_r)\ ,\end{aligned}$$

where we have used the Feynman technique and

$$I_{ij}(t_r) = \int\frac{d\omega}{(2\pi)^{1/2}}\exp(-i\omega t_r)I_{ij}(\omega)\ .$$

Physically, it means that the gravitational wave produced is proportional to the second derivative of the quadrupole moment of energy density.

5.4 Binary system

Consider the gravitational wave emitted by a binary system composed by two stars of equal mass M orbiting around their center of mass R in the plane $x^1 - x^2$. Using the Newtonian-Keplerian approximation, we can find the angular frequency of the orbit

$$\Omega = \left(\frac{G_NM}{4R^3}\right)^{1/2}\ .$$

Proof. Equating the centrigual force with the gravitational one

$$\frac{G_N M}{(2R)^2} = \frac{Mv^2}{R} \ , \quad v = \Big(\frac{G_N M}{4R}\Big)^{1/2} \ .$$

The frequency is given by

$$\Omega = \frac{2\pi}{T} = \frac{2\pi v}{2\pi R} = \Big(\frac{G_N M}{4R^3}\Big)^{1/2} \ .$$

q.e.d.

The explicit path is given by

$$x_a^1 = R\cos(\Omega t) \ , \quad x_a^2 = R\sin(\Omega t) \ , \quad x_b^1 = -R\cos(\Omega t) \ , \quad x_b^2 = -R\sin(\Omega t) \ ,$$

where a and b are different stars. The energy density is given by

$$T^{00}(t,\underline{x}) = M\delta(x^3)(\delta(x^1 - R\cos(\Omega t))\delta(x^2 - R\sin(\Omega t)) + \delta(x^1 + R\cos(\Omega t))\delta(x^2 + R\sin(\Omega t))) \ .$$

The quadrupole tensor is

$$I_{ij} = \begin{bmatrix} MR^2(1+\cos(2\Omega t)) & MR^2\sin(2\Omega t) & 0 \\ MR^2\sin(2\Omega t) & MR^2(1-\cos(2\Omega t)) & 0 \\ 0 & 0 & 0 \end{bmatrix} \ . \tag{5.2}$$

Proof. In fact

$$\begin{aligned} I_{11} &= \int d^3y \ T^{00}(t,\underline{y}) y_1 y_1 \\ &= \int d^3y \ M\delta^3(y^3)\Big(\delta(y^1 - R\cos(\Omega t))\delta(y^2 - R\sin(\Omega t)) \\ &\qquad + \delta(y^1 + R\cos(\Omega t))\delta(y^2 + R\sin(\Omega t))\Big) y_1 y_1 \\ &= M\Big(\int dy^1 \ y_1 y_1 \delta(y_1 - R\cos(\Omega t)) \int dy_2 \ \delta(y_2 - R\sin(\Omega t)) \int dy_3 \ \delta(y_3) \\ &\qquad + \int dy^1 \ y_1 y_1 \delta(y_1 + R\cos(\Omega t)) \int dy_2 \ \delta(y_2 + R\sin(\Omega t)) \int dy_3 \ \delta(y_3)\Big) \\ &= 2MR^2\cos^2(\Omega t) = MR^2(1+\cos(2\Omega t)) \ , \end{aligned}$$

$$
\begin{aligned}
I_{12} = I_{21} &= \int d^3y\ T^{00}(t,\underline{y})y_1y_2 \\
&= \int d^3y\ M\delta^3(y^3)\Big(\delta(y^1 - R\cos(\Omega t))\delta(y^2 - R\sin(\Omega t)) \\
&\qquad + \delta(y^1 + R\cos(\Omega t))\delta(y^2 + R\sin(\Omega t))\Big)y_1y_2 \\
&= M\Big(\int dy^1\ y_1\delta(y_1 - R\cos(\Omega t))\int dy_2\ y_2\delta(y_2 - R\sin(\Omega t))\int dy_3\ \delta(y_3) \\
&\qquad + \int dy^1\ y_1\delta(y_1 + R\cos(\Omega t))\int dy_2\ y_2\delta(y_2 + R\sin(\Omega t))\int dy_3\ \delta(y_3)\Big) \\
&= 2MR^2\sin(\Omega t)\cos(\Omega t) = MR^2\sin(2\Omega t)\ ,
\end{aligned}
$$

$$
\begin{aligned}
I_{22} &= \int d^3y\ T^{00}(t,\underline{y})y_2y_2 \\
&= \int d^3y\ M\delta^3(y^3)\Big(\delta(y^1 - R\cos(\Omega t))\delta(y^2 - R\sin(\Omega t)) \\
&\qquad + \delta(y^1 + R\cos(\Omega t))\delta(y^2 + R\sin(\Omega t))\Big)y_2y_2 \\
&= M\Big(\int dy^1\ \delta(y_1 - R\cos(\Omega t))\int dy_2\ y_2y_2\delta(y_2 - R\sin(\Omega t))\int dy_3\ \delta(y_3) \\
&\qquad + \int dy^1\ \delta(y_1 + R\cos(\Omega t))\int dy_2\ y_2y_2\delta(y_2 + R\sin(\Omega t))\int dy_3\ \delta(y_3)\Big) \\
&= 2MR^2\sin^2(\Omega t) = MR^2(1 - \cos(2\Omega t))\ .
\end{aligned}
$$

q.e.d.

The gravitational wave energy-momentum tensor in the transverse traceless gauge is given by

$$t_{\mu\nu} = \frac{1}{32\pi G_N}\langle\partial_\mu h^{TT\alpha\beta}\partial_\nu h^{TT}_{\alpha\beta}\rangle\ .$$

Since it is in the transverse traceless gauge, we need to introduce the reduced quadrupole moment

$$I^{TT}_{ij} = I_{ij} - \frac{1}{3}\delta_{ij}\delta^{kl}I_{kl}\ ,$$

and the quadrupole formula becomes

$$h^{TT}_{ij} = \frac{2G_N}{r}\frac{d^2}{dt^2}I^{TT}_{ij}(t_r)\ .$$

The power emitted by the system is given by

$$P = -\frac{G_N}{5}\langle\frac{d^3}{dt^3}I^{TT}_{ij}\frac{d^3}{dt^3}I^{TTij}\rangle\ .$$

In our binary merge case, we have the reduced quadrupole moment to be

$$I_{ij}^{TT} = \frac{MR^2}{3}\begin{bmatrix} 1+3\cos(2\Omega t) & 3\sin(2\Omega t) & 0 \\ 3\sin(2\Omega t) & 1-3\cos(2\Omega t) & 0 \\ 0 & 0 & -2 \end{bmatrix} .$$

Proof. In fact, the trace of (5.2) is

$$\delta^{ij} I_{ij} = 2MR^2 .$$

The reduced quadrupole moment is

$$\begin{aligned} I_{ij}^{TT} &= I_{ij} - \frac{1}{3}\delta_{ij}\delta^{kl} I_{kl} \\ &= \begin{bmatrix} MR^2(1+\cos(2\Omega t)) & MR^2\sin(2\Omega t) & 0 \\ MR^2\sin(2\Omega t) & MR^2(1-\cos(2\Omega t)) & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{2}{3}MR^2 \\ &= \frac{MR^2}{3}\begin{bmatrix} 1+3\cos(2\Omega t) & 3\sin(2\Omega t) & 0 \\ 3\sin(2\Omega t) & 1-3\cos(2\Omega t) & 0 \\ 0 & 0 & -2 \end{bmatrix} \end{aligned}$$

q.e.d.

Finally, the power radiated is

$$P = -\frac{2}{5}\frac{G_N^4 M^5}{R^5} .$$

Proof. The third derivative is

$$8MR^2\Omega^3 \begin{bmatrix} \sin(2\Omega t) & -\cos(2\Omega t) & 0 \\ -\cos(2\Omega t) & -\sin(2\Omega t) & 0 \\ 0 & 0 & 0 \end{bmatrix} .$$

Therefore, the power is

$$P = -\frac{128}{5} G_N M^2 R^4 \Omega^6 = -\frac{2}{5}\frac{G_N^4 M^5}{R^5} .$$

q.e.d.

5.5 BOH

We can approximate on the center of mass $\underline{r}$, so that $h^T_{ij}(t,r)$. Outside the source, we can find the solution using the Green function of the d'Alambertian

$$G^T_{ijkl}(t,r,t',r') \sim \frac{\delta(t'-t+r)}{r} \ ,$$

hence

$$h^T_{ij}(t,r) = 16\pi G_N \int d^3r' \ G^T_{ijkl}(t,r,t',r') T^{kl}(t',r') \ ,$$

such that $G_{ijkl} = G_{jikl}$ and $G^i{}_{ikl} = 0$. Noticing that

$$T^{ij}(t',r) \simeq \rho(t',r')\delta\dot{r}^i(t')\dot{r}^j(t') \ ,$$

$$\int d^3r' \ T^{ij}(t',r') \simeq \int d^3x \ \rho(t',\underline{r}+\underline{x})\dot{x}^i\dot{x}^j \simeq \frac{d^2}{dt^2}\int d^3x \ \rho(t',\underline{r}+\underline{x})x^ix^j \ ,$$

where we have neglected $\dot{\rho}$ and $\ddot{x}^i$, we define the traceless quadrupole

$$I_{ij} \simeq \int d^3x \ \rho(\underline{r}+\underline{x})\Big(x^ix^j - \frac{1}{3}\delta_{ij}x^2\Big)$$

and we find the quadrupole formula

$$h^T_{ij}(t,r) \simeq \frac{16\pi G_N}{r}\ddot{I}_{ij}(t-r) \ ,$$

where $t-r$ is the retarded.

Since gravity is described by the metric, there is no force and therefore no energy. However, we can identify the energy of a gravitational wave by the work done on matter. In our approximation, this is

$$\dot{E} \sim G_N \dddot{I}_{ij}\dddot{I}^{ij} \ .$$

Consider a binary system of orbital radius R and velocity

$$v^2 \sim \frac{G_N M}{R} \ .$$

The strain is

$$a \sim \frac{\Delta l}{l} \sim \frac{G_N M v^2}{r} \sim \frac{G_N^2 M^2}{rR}$$

Furthermore, the frequency is

$$\omega \sim \frac{v}{R} \sim \sqrt{\frac{G_N M}{R^3}} \ .$$

5.6 Detection

We can detect gravitational waves by means of a Michelson-Morley interferometer. Assuming $v \ll c$ and $t \sim l_0 \ll l_0/v$, we find the maximum travel time

$$\Delta t \simeq 2(l_0(1+a)^{1/2} - l_0(1-a)^{1/2}) \simeq a l_0 \ .$$

Looking at the interference pattern, we find that it is proportional to the strain

$$a \sim \frac{\Delta l}{l} \simeq \frac{\Delta l}{l} \ .$$

Part III

Black holes

Chapter 6

Exterior Schwarzschild metric

It was the 1916 when Schwarzschild, during the World War I, solved the Einstein field equations and found the famous Schwarzschild metric, which after lead to concepts as Schwarzschild radius and black holes. In this chapter, after some reasonable assumptions of static spherical symmetry of the source, we will derive a suitable metric to mathematically describe the stars and find the corresponding Einstein field equations. Furthermore, we will solve to find the metric we are looking for.

6.1 Symmetries

Consider a spherical symmetric source. We are interested in the region outside it, where $T^{\mu\nu} = 0$, which leads to the vacuum Einstein field equations

$$G_{\mu\nu} = 0 \ .$$

It is possible to prove that they correspond to

$$R_{\mu\nu} = 0 \ .$$

Proof. In fact, we take the trace

$$g^{\mu\nu} G_{\mu\nu} = g^{\mu\nu}(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R) = R - 2R = 0 \ ,$$

hence,

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \underbrace{R}_{0} = R_{\mu\nu} \ .$$

q.e.d.

The solution will depend on free parameters that can be derived in two different way: some sort of continuity with the interior solution or the Newtonian limit. We will study the second case.

The metric has 10 independent components, but it can be much simplified by studying the symmetries of the system. Staticity entails the existence of a time-like Killing vectors K_t; spherical symmetry entails the existence of three rotational space-like Killing vectors $K_i = d/d\theta_i$ with $i = 1, 2, 3$ such that they satisfy the Lie algebra $[K_i, K_j] = \epsilon_{ijk} K_k$. Space-like Killing vectors are conserved in time, therefore $[K_t, K_i] = 0$. Concerning radial component, there is no Killing vector associated to it. Accordingly, the metric depends only on functions of (coordinate) radius $A(r)$, $B(r)$ and $C(r)$ to be determined by the Einstein field equations. Adopting the polar coordinates (t, r, θ, ϕ), the metric explicitly looks like

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + C(r)(d\theta^2 + \sin^2\theta d\phi^2) = -A(r)dt^2 + B(r)dr^2 + C(r)d\Omega^2 \ .$$

It is possible to exploit the possibility to rescale the r in such a way that the area of a sphere is given by the formula

$$S(r) = \int C(r)d\Omega^2 = r^2 \int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\phi = 4\pi r^2 \ .$$

which implies that $C(r) = r^2$. Therefore, the metric becomes

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega^2 \ .$$

Notice that the proper radius is instead

$$R(r) = \int_0^r dr \ \sqrt{g_{rr}} = \int_0^r dr \ \sqrt{B(r)} \ . \tag{6.1}$$

In order to find a chart, we need to take into account also domains of the coordinates. We reasonably set $t \in \mathbb{R}$, $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi[$. However, we cannot say anything about r. It is import to highlight that the metric is not the only unknown, but also the manifold needs to be found. In fact, it possible that this chart does not cover the whole of it.

6.2 Computations

In matrix notation, the metric takes the form

$$g = \begin{bmatrix} -A(r) & 0 & 0 & 0 \\ 0 & B(r) & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} ,$$

.

The non-vanishing Christoffel symbols are

$$\Gamma^0_{01} = \frac{0.5\frac{d}{dr}A(r)}{A(r)} \ , \quad \Gamma^0_{10} = \frac{0.5\frac{d}{dr}A(r)}{A(r)} \ ,$$
$$\Gamma^1_{00} = \frac{0.5\frac{d}{dr}A(r)}{B(r)} \ , \quad \Gamma^1_{11} = \frac{0.5\frac{d}{dr}B(r)}{B(r)} \ ,$$
$$\Gamma^1_{22} = -\frac{1.0r}{B(r)} \ , \quad \Gamma^1_{33} = -\frac{1.0r\sin^2(\theta)}{B(r)} \ ,$$
$$\Gamma^2_{12} = \frac{1.0}{r} \ , \quad \Gamma^2_{21} = \frac{1.0}{r} \ ,$$
$$\Gamma^2_{33} = -0.5\sin(2\theta) \ , \quad \Gamma^3_{13} = \frac{1.0}{r} \ ,$$
$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ , \quad \Gamma^3_{31} = \frac{1.0}{r} \ ,$$
$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ , \quad .$$

The non-vanishing Ricci tensor components are

$$R_{00} = \frac{0.5\frac{d^2}{dr^2}A(r)}{B(r)} - \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A(r)B(r)} + \frac{1.0\frac{d}{dr}A(r)}{rB(r)} \ ,$$
$$R_{11} = -\frac{0.5\frac{d^2}{dr^2}A(r)}{A(r)} + \frac{0.25\frac{d}{dr}A(r)\frac{d}{dr}B(r)}{A(r)B(r)} + \frac{0.25\left(\frac{d}{dr}A(r)\right)^2}{A^2(r)} + \frac{1.0\frac{d}{dr}B(r)}{rB(r)} \ ,$$
$$R_{22} = \frac{0.5r\frac{d}{dr}B(r)}{B^2(r)} - \frac{0.5r\frac{d}{dr}A(r)}{A(r)B(r)} + 1.0 - \frac{1.0}{B(r)} \ ,$$
$$R_{33} = \frac{\left(0.5rA(r)\frac{d}{dr}B(r) - 0.5rB(r)\frac{d}{dr}A(r) + 1.0A(r)B^2(r) - 1.0A(r)B(r)\right)\sin^2(\theta)}{A(r)B^2(r)} \ ,$$

.

6.3 Schwarzschild metric

The Einstein field equations reduce to a set of differential equations in $A(r)$ and $B(r)$ to solve

$$R_{00} = -\frac{A''}{2B} + \frac{(A')^2}{4AB} + \frac{A'B'}{4B^2} - \frac{A'}{rB} = 0 \ ,$$
$$R_{11} = -\frac{A''}{2A} + \frac{(A')^2}{4A^2} + \frac{B'A'}{4AB} + \frac{B'}{rB} = 0 \ ,$$
$$R_{22} = 1 - \frac{1}{B} + \frac{rB'}{2B^2} - \frac{rA'}{2AB} = 0 \ ,$$
$$R_{33} = \sin^2\theta R_{22} = 0 \ .$$

Notice that a linear combination of the first two gives

$$
\begin{aligned}
0 &= \frac{B}{A} R_{00} - R_{11} \\
&= -\cancel{\frac{B}{A}\frac{A''}{2B}} + \cancel{\frac{B}{A}\frac{(A')^2}{4AB}} + \cancel{\frac{B}{A}\frac{A'B'}{4B^2}} - \frac{B}{A}\frac{A'}{rB} + \cancel{\frac{A''}{2A}} - \cancel{\frac{(A')^2}{4A^2}} - \cancel{\frac{B'A'}{4AB}} - \frac{B'}{rB} \\
&= -\frac{A'}{rA} - \frac{B'}{rB} \ .
\end{aligned}
$$

Hence, we find

$$
0 = A'B + B'A = (AB)' \ , \quad AB = const \ ,
$$

which implies, after a time rescaling to set the constant to unity, that

$$
A = B^{-1} \ .
$$

Therefore, the third equation becomes

$$
\begin{aligned}
0 &= 1 - \frac{1}{B} + \frac{rB'}{2B^2} - \frac{rA'}{2AB} \\
&= 1 - A - \frac{rA'}{2} - \frac{rA'}{2} = 1 - A - rA'
\end{aligned}
$$

and the solution is

$$
1 - A = r\frac{dA}{dr} \ , \quad \frac{dA}{1-A} = \frac{dr}{r} \ ,
$$

$$
-\ln(1-A) = \ln r + C \ , \quad 1 - A = \frac{\exp(-C)}{r} \ ,
$$

$$
A = 1 - \frac{K}{r} \ ,
$$

where we have called $K = \exp(C)$. As we said before, the constant K is a free parameter (integration constant) that can be found by considering the Newtonian limit. In fact, noticing that we can write

$$
A = g_{00} = 1 + \frac{K}{r} = \eta_{00} + h_{00} \ ,
$$

and recalling that

$$
h_{00} = -2V_N = \frac{2G_N M}{r} \ ,
$$

we find

$$
h_{00} = \frac{2G_N M}{r} = \frac{K}{r} \ , \quad K = 2G_N M \ .
$$

At last, we have obtained the (exterior) Schwarzschild metric

$$
g_{\mu\nu} = \begin{bmatrix} -(1 - R_H/r) & 0 & 0 & 0 \\ 0 & (1 - R_H/r)^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2\theta \end{bmatrix} \ ,
$$

$$ds^2 = -\Big(1 - R_H/r\Big)dt^2 + \Big(1 - R_H/r\Big)^{-1} dr^2 + r^2 d\Omega^2 \ .$$

where we have used quantity $R_H = 2G_N M$, called the Schwarzschild radius. This quantity is important in the Schwarzschild spacetime. Notice that at spatial infinity $r \gg R_H$ or $K \ll 1$, we have an asymptotic flatness and we recover the Minkowski metric. This static observers are inertial and they differ from it the more they approach R_H.

The deviation of the Schwarzschild potential from the Newtonian one can be found by looking at the harmonic gauge. In fact, t satisfies the Donder gauge $\Box t = 0$, but the spatial coordinates not. However, with a change of coordinates into $\tilde{r} = r - G_N M$, it is possbile to satisfy the gauge. In the Newtonian limit for $r \gg G_N M$, we have the expansion for $x = G_N M/\tilde{r} \ll 1$ and $\tilde{r} \simeq r$

$$\begin{aligned} V &= \frac{1 + g_{tt}}{2} = -\frac{G_N M}{r} = -\frac{G_N M}{\tilde{r} + G_N M} = -\frac{G_N M}{\tilde{r} + G_N M} = -\frac{x}{1+x} \simeq -x + x^2 - x^3 \\ &= -\frac{G_N M}{\tilde{r}} + \frac{G_N^2 M^2}{\tilde{r}^2} - \frac{G_N^3 M^3}{\tilde{r}^3} \ , \end{aligned}$$

where we have used the expansion

$$\frac{x}{1+x} \simeq x - x^2 + x^3 + O(x^4) \ .$$

A plot that shows the difference between the Newtonian and the Schwarzschild potential is in Figure 6.1.

6.4 Birkhoff theorem

There is a theorem, due to Birkhoff, that helps us understand why we do not have $A(t,r)$ or $B(t,r)$.

Theorem 6.1 (Birkhoff)
A spherically symmetric metric which solves the vacuum Einstein field equations is static.

Proof. Maybe in the future. q.e.d.

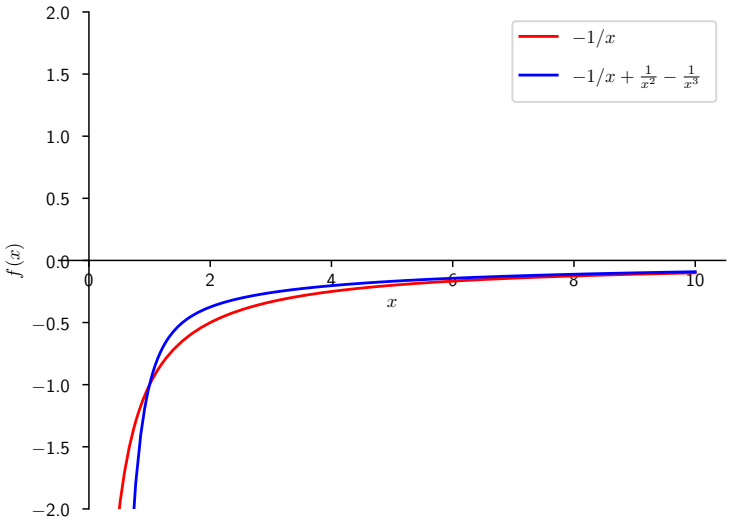

Figure 6.1: Comparison between the Newtonian (in red) and the Schwarzschild (in blue) potential.

Chapter 7

Physics in Schwarzschild metric

7.1 Radial geodesics

Studying the proper radius of a sphere is not very useful

$$R(r) = r\sqrt{1-\frac{r_H}{r}} + \frac{r_H}{2}\ln\Big(\frac{2r}{r_H}\Big(1+\sqrt{1-\frac{r_H}{r}}-1\Big)\Big) \ .$$

Therefore, we

Since the Schwarzschild metric does not carry dependence of the source size, we can suppose that $r_S \ll R_H$ and the metric becomes for $r \to R_H$, $g_{tt} = 0$ and $g_{rr} \to \infty$. Furthermore, the signs of g_{tt} and g_{rr} change across $r = R_H$. The Killing vector $\underline{K}_t$ is defined everywhere but $\underline{K}_r$ is not defined. This means that for inside and outside R_H, we maybe need two different manifolds, with the same coordinates (t, r, θ, ϕ) but different domains $r \in (0, R_H)$ for inside and $r \in (R_H, \infty)$ for outside. Notice that we need to find a chart that covers R_H. Therefore, we need to study a test particle that travels from outside to inside.

Consider the action of an infalling massive test particle

$$S = m\int_0^\tau d\tau\ \sqrt{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu} = m\int_0^\tau d\tau\ \sqrt{2T} \ ,$$

where

$$2T = (1-\frac{2G_N M}{r}\Big)\dot{t}^2 - (1-\frac{2G_N M}{r}\Big)^{-1}\dot{r}^2 - r^2\dot{\theta}^2 - r^2\sin^2\theta\dot{\phi}^2 = 1 \ .$$

The Euler-Lagrange equations are

$$\frac{d}{d\tau}\Big(\frac{\partial T}{\partial \dot{x}^\mu}\Big) - \frac{\partial T}{\partial x^\mu} = 0 \ ,$$

because

$$\delta S = m\delta\int_0^\tau d\tau\ \sqrt{2T} = m\int_0^\tau d\tau\frac{\delta T}{\sqrt{2T}} = m\delta\int_0^\tau d\tau\ T \ .$$

Since the Lagrangian does not depend on t and ϕ, we can define two conserved quantities: energy

$$E = \frac{\partial T}{\partial \dot{t}} = (1 - \frac{2G_N M}{r})\dot{t} \ ,$$

and angular momentum

$$L = \frac{\partial T}{\partial \dot{\phi}} = -r^2 \sin^2\theta \dot{\phi} \ .$$

Notice that the energy reduces to 1 in the weak-field limit. By means of the Killing vectors $\underline{K}_i$, we can rotate the reference frame in order to have $\theta = \pi/2$. In fact, the Euler-Lagrange equation for θ are identically satisfied

$$0 = \frac{d}{d\tau}\Big(\frac{\partial T}{\partial \dot{\theta}}\Big) - \frac{\partial T}{\partial \theta} = \ddot{\theta} - \sin\theta\cos\theta\dot{\phi}^2 \ .$$

Now, consider the simple case $\phi = 0$ of radially in-falling particles. The energy equation becomes

$$2T = (1 - \frac{2G_N M}{r}\Big)^{-1}(E^2 - \dot{r}^2) = 1 \ ,$$

where we have set $\theta = \pi/2$ and $\phi = 0$. We solve this equation and we find

$$\dot{t} = (1 - \frac{2G_N M}{r}\Big)^{-1} E \ ,$$

which implies that

$$\frac{dt}{d\tau} \sim \frac{1}{r - R_H} \ .$$

Physically, it means that an asymptotically inertial observer at spatial infinity will take an infinite amount of coordinate time to see a particle falling radially. Now, we invert the energy equation

$$\dot{r}^2 = \frac{2G_N M}{r} + (E^2 - 1) \ ,$$

and we derive it

$$\ddot{r} = -\frac{G_N M}{r^2} \ .$$

It is similar to the Newton law, but it is not valid for $r > 0$. It can be solved by an ansatz

$$r = r_0\Big(\frac{\tau}{\tau_0}\Big)^{\alpha} \ ,$$

with initial condition $r(\tau_0) = \infty$ and $\dot{r}(\tau) = 0$. α turns out to be $2/3$.

Proof. In fact,

$$\ddot{r} = r_0\tau_0^{-\alpha}\alpha(\alpha - 1)\tau^{\alpha-2} = -\frac{G_N M}{r^2} = -G_N M r_0^{-2}\tau_0^{2\alpha}\tau^{-2\alpha} \ .$$

q.e.d.

Notice that, if we have two radially in-falling particles, starting at the same distance r_0 with the same velocity, they will measure the same proper time. Their arc distance is proportional to $r = r(\tau)$. Therefore, the equation will measure the tidal forces perpendicular to the falling line, since it is proportional to the geodesics. These forces diverges for $r \to 0$. By making a variation of the equation, we find

$$\delta\ddot{r} \simeq \frac{G_N M}{r^3}\delta r \ .$$

This phenomenom is called spaghettification.

The geodesics for massless particle can be found by setting $ds^2 = d\theta^2 = d\phi^2 = 0$. Hence

$$\frac{dr}{dt} = \pm\Big(1 - \frac{2G_N M}{r}\Big) \ .$$

where the plus means that it is falling and minus it is climbing.

7.2 Gravitational red-shift

Consider a radial geodesic for photons, i.e. $T = 0$. Since the metric varies significally from Minkowski only near $r \simeq R_H$, we cannot put an experimental apparatus there. However, we can look at the signal that comes from an in-falling probe, starting from far away.

Since there exists a time-like Killing vector $\underline{K}_t$, its energy is conserved along a geodesic

$$E = -K_t^\mu u_\mu \ ,$$

where u^μ is the 4-velocity of a particle along the geodesic, parametrised by λ.

Proof. In fact

$$\frac{dE}{d\lambda} = -u^\nu \nabla_\nu (K_t^\mu u_\mu) = -K_t^\mu \underbrace{u^\nu \nabla_\nu u_\mu}_{0} + u^\nu u^\mu \nabla_\nu K_{t\mu} = 0 \ ,$$

where we have used $d/d\lambda = u^\nu \nabla_\nu$, the geodesic equation and the Killing equation. q.e.d.

Therefore, the 4-velocity for a static observer U^μ is

$$U^\mu = \frac{K_t^\mu}{\sqrt{-K_t^\nu K_{t\nu}}} \ .$$

Notice that U^μ is normalised at $U^\mu U_\mu = -1$ whereas $\underline{K}_t$ not. The energy measured by this static observer is

$$\omega = -U^\mu u_\mu = -\frac{K_t^\mu u_\mu}{\sqrt{-K_t^\nu K_{t\nu}}} = \frac{E}{|K_t|} \ .$$

Now consider a photon which crosses two different static observer at points r_1 and r_2. They will measure photon's energies

$$\frac{\omega_1}{\omega_2} = \frac{|K_t(r_2)|}{|K_t(r_1)|} \ .$$

In particular, at constant r, we have

$$U^\mu = (\dot{t}(r), 0, 0, 0) = \dot{t} K_t^\mu \ ,$$

and

$$\frac{\omega_1}{\omega_2} = \frac{\dot{t}(r_1)}{\dot{t}(r_2)} = \sqrt{\frac{1 - R_H/r_1}{1 - R_H/r_1}} < 1 \ ,$$

where we assumed $r_1 > r_2$. Physically, it means that the photon seems to lose energy as it climbs. If we add also the relativistic Doeppler effect, due to the velocity v_2, we have the complete formula

$$\omega_1 \simeq \omega_s \sqrt{\frac{1 - v_2}{1 + v_2}} \sqrt{\frac{1 - R_H/r_1}{1 - R_H/r_1}} \ .$$

The gravitational effect depends on the geometry, whereas the Doeppler effect depends on the initial conditions.

Consider a probe sent radially towards the source and a static observer at $r_0 \gg R_H$. The probe emits signals of frequency ω_s at fixed intervals $\Delta\tau_s$, constant since the probe is an inertial frame. The points of emission are $S_n = (\tau_n = n\Delta\tau_s, r_n)$, whereas the point of detection are $O_n = (t_n, r_0)$. The equations of an in-falling probe are

$$\frac{dr_s}{d\tau} = -\sqrt{\frac{R_H}{r_s} - 1 + E^2} \ , \quad \frac{d^2 r_s}{d\tau^2} = -\frac{R_H}{2r_s^2} \ ,$$

where $E^2 = 1 - R_H/r_0 \simeq 1$. Therefore

$$\frac{dr_s}{d\tau} \simeq -sqrt\frac{R_H}{r_s} \ , \quad \frac{dt}{d\tau} = \left(1 - \frac{R_H}{r_s}\right)^{-1} E \simeq \left(1 - \frac{R_H}{r_s}\right)^{-1} \ .$$

The out-going signal is

$$\frac{dr_\gamma}{dt} = 1 - \frac{2G_N M}{r_\gamma} \ , \quad \omega_n = \omega_n' \sqrt{\frac{1 - R_H/r_n}{1 - R_H/r_0}} \simeq \omega_n' \sqrt{1 - \frac{R_H}{r_n}} \ ,$$

where ω_n' is the frequency measured by a static observer in r_n. The interval of detection is composed by two phenomena: in-falling probe $\Delta t_n^{(1)}$ and the difference in travel time $\Delta t_n^{(2)}$. The former is given by

$$\Delta t_n^{(1)} = \frac{\Delta\tau_s}{1 - R_H/r_n} \ .$$

For the latter, the radial difference is

$$\delta r_n = r_n - r_{n-1} = \int_{r_{n-1}}^{r_n} d\tau \frac{dr_s}{d\tau} \simeq -\int_{\tau_{n-1}}^{\tau_n} d\tau \sqrt{\frac{R_H}{r_s(\tau)}} \simeq \sqrt{\frac{R_H}{r_s(\tau)}} \Delta\tau_s \ ,$$

hence

$$\begin{aligned}\Delta t_n^{(2)} = t_n - t_{n-1} &= \int_{r_n}^{r_0} dr_\gamma \frac{dt}{dr_\gamma} - \int_{r_{n-1}}^{r_0} dr_\gamma \frac{dt}{dr_\gamma} = -\int_{r_{n-1}}^{r_n} dr_\gamma \frac{dt}{dr_\gamma} \\ &= -\int_{r_{n-1}}^{r_n} dr_\gamma \Big(1 - \frac{R_H}{r_\gamma}\Big)^{-1} \simeq -\Big(1 - \frac{R_H}{r_\gamma}\Big)^{-1} \Delta r_n \simeq \sqrt{\frac{R_H}{r_n}} \frac{\Delta\tau_s}{1 - R_H/r_n} \ .\end{aligned}$$

Puttin together, we have

$$\Delta t_n = \Delta t_n^{(1)} + \Delta t_n^{(2)} \simeq \Big(1 + \sqrt{\frac{R_H}{r_n}}\Big) \frac{\Delta\tau_s}{1 - R_H/r_n} \ .$$

Physically, it means that the observer will receive less pulse per unit of time.

If the probe remained at fixed radius, we would have only the first term

$$\Delta_n^{(1)} = \frac{\Delta T_s}{\sqrt{1 - R_H/r_n}} \ ,$$

where we have used $\Delta\tau_s = \Delta T_s$. If we interpret $\Delta T_s = 1/\omega_2$, we recover the gravitational red-shift for $r_0 = r_1 \gg R_H$.

Using the total red-shift

$$\omega_n \simeq \omega_s \sqrt{\frac{1 - v_n}{1 + v_n}} \sqrt{1 - \frac{R_H}{r_n}} \ ,$$

we can estimate the velocity with respect to the local static observer

$$v_n = \frac{dr_s}{dT}\Big|_{t_n} = \frac{dr_s}{d\tau}\frac{d\tau}{dt}\frac{dt}{dT}\Big|_{t_n} \simeq \sqrt{\frac{R_H}{r_n}\Big(1 - \frac{R_H}{r_n}\Big)} \ .$$

In particular, for $r_s \to R_H$, we have $v_n \to 0$ and the observed frequency vanishes. Therefore, there is a radius $r_s > R_H$ such that $\omega_n < \omega_c$, i.e. there is a black-out.

7.3 Black holes

The gravitational red-shift implies that a photon would spend all of its energy to escape. Therefore, the Schwarzschild sphere is called the event horizon. No signal can escape from it, in fact for $r > R_H$ there exists both out-going and in-going light conse, for $r \leq R_H$ there exists only in-going (out-going will be stuck on $r = R_H$ or it will contract). The inside is called a black hole. See Table (7.1). It is important to highlight that the real singularity is at $r = 0$ whereas for the event horizons there is only a coordinate singularity. Black hole can be formed by a collapse of a sphere of dust, via neutron stars or white dwarf.

Black hole	Charged	Rotation
Schwarzschild	no	no
Reissner-Nordstroem	yes	no
Kerr	no	yes
Kerr-Newman	yes	yes

Table 7.1: A table of all mathematically known black holes.

7.4 Orbits

Notice that if we multiply by $m/2$, the equation of the energy

$$\dot{r}^2 = \frac{2G_N M}{r} + E^2 - 1 \ ,$$

we find a Newtonian-like expression

$$\frac{m}{2}\dot{r}^2 - \frac{G_N m M}{r} = \frac{m}{2}(E^2 - 1) \ ,$$

where the Newtonian energy is

$$E_N = \frac{m}{2}(E^2 - 1) \ .$$

If $E_N < 0$ or $E^2 < 1$, the orbit is bounded. If $E_N = 0$ or $E^2 = 1$, the orbit is radially in-falling. If $E_N > 0$ or $E^2 > 1$, the orbit is unbounded.

The equation of radial motion of a test particle is

$$\frac{1}{2}\left(\frac{dr}{d\tau}\right)^2 + V(r) = \frac{E^2}{2} = \mathcal{E} \ ,$$

where the potential is

$$V(r) = \frac{\epsilon}{2}\left(1 - \frac{2G_N M}{r}\right) + \frac{L^2}{2r^2} - \gamma\frac{G_N M L^2}{r^3} \ ,$$

with $\gamma = 1$ for general relativity and $\gamma = 0$ for Newtonian gravity. $\epsilon = 1$ for massive particles and $\epsilon = 0$ for massless ones. Now, we delve the solution of this 1-dimensional equation.

Consider circular orbit, whose radius is given by

$$0 = \frac{dV}{dr}\bigg|_{r_c} \sim \epsilon G_N M r_c^2 - L^2 r_c + 3\gamma G_N M L^2 \ .$$

This critical radius is stable if they are a minimum, otherwise it is unstable. In Newtonian gravity for $\gamma = 0$, we have for massless particles no circular orbits. Instead, for massive particles, we have

$$r_c = \frac{L^2}{G_N M} \ .$$

In general relativity $\gamma = 1$, we have for massless particles

$$r_c = 3G_N M \ ,$$

which is unstable. Instead, for massive particles, we have

$$r_\pm = \frac{L^2}{2G_N M}\Big(1 \pm \sqrt{1 - \frac{12G_N^2 M^2}{L^2}}\Big) \ .$$

Hence, for $L > \sqrt{12}G_N M$, the orbit is unstable for r_- and stable for r_+. For large L, r_- recovers the massless case and r_+ recovers the Newtonian case. For small L, they coincide for $L = \sqrt{12}G_N M$ with radius $r_c = 6G_N M$. For $L < \sqrt{12}G_N M$, no circular orbits are possible.

Non-circular orbits can be viewed as ellipses that precess with $r = r(\phi)$. Substituting $L = r^2 d\phi/dr$ for $\theta = \pi/2$, we have

$$\Big(\frac{dr}{d\phi}\Big)^2 = \frac{E^2 - \epsilon}{L^2} r^4 + \frac{2\epsilon G_N M}{L^2} r^3 - r^2 + 2\gamma G_N M r \ ,$$

which becomes for $x = L^2/G_N M r$

$$\Big(\frac{x}{\phi}\Big)^2 = \frac{L^2(E^2 - \epsilon)}{G_N^2 M^2} + 2\epsilon x - x^2 + \gamma \frac{2G_N^2 M^2}{L^2} x^3 \ ,$$

$$\frac{d^2 x}{d\phi^2} = \epsilon - x + \gamma \frac{3G_N^2 M^2}{L^2} x^2 \ .$$

Defining the dimensionless quantities

$$\alpha = \frac{G_N M}{L} \ , \quad \beta = \frac{G_N M}{EL} = \frac{\alpha}{E} \ , \quad \rho = \frac{2G_N^2 M^2}{L^2} = 2\alpha^2 \ ,$$

we obtain

$$\Big(\frac{x}{\phi}\Big)^2 = \Big(\frac{1}{\beta^2} - \frac{\epsilon}{\alpha^2}\Big) + 2\epsilon x - x^2 + \gamma \rho x^3 \ ,$$

$$\frac{d^2 x}{d\phi^2} = \epsilon - x + \frac{3}{2}\gamma \rho x^2 \ ,$$

For Newtonian gravity $\gamma = 0$, we have

$$\Big(\frac{x}{\phi}\Big)^2 = \Big(\frac{1}{\beta^2} - \frac{\epsilon}{\alpha^2}\Big) = -(x - x_+)(x - x_-) \ ,$$

$$\frac{d^2 x}{d\phi^2} = \epsilon - x \ .$$

Calling $x_\mp = \epsilon \mp k$, with

$$k = \sqrt{\frac{1}{\beta^2} - \frac{\epsilon}{\alpha^2} + \epsilon^2} \ .$$

The general solution is

$$x(\phi) = x_- + (x_+ - x_-)\sin^2(\phi/2 + \delta) = \epsilon(x_- - \epsilon)\cos(\phi + 2\delta) \ ,$$

where δ is an integration constant. The eccentricity is given by

$$e = \frac{x_+ - x_-}{x_+ + x_-} = \frac{k}{\epsilon} \ .$$

For massive particles, we have: if $x_- > 0$ and $x_- \neq x_+$ then the orbit is an ellipse with x_- perihelion and x_+ aphelion; if $x_- = x_+$ or $k = 0$ then the orbit is circular; if $x_- = 0$ or $k = 1$ then the orbit is a parabola and x_2 is the vertex; if $x_1 < 0$ or $k > 1$ then the orbit is an hyperbola and x_2 is the vertex. For massless particles, $x \in [-k, k]$ and the orbit is unbounded.

In general relativity $\gamma = 1$, we have

$$\left(\frac{dx}{d\phi}\right)^2 = \rho(x - x_1)(x - x_2)(x - x_3) \ ,$$

with $x_1 + x_2 + x_3 = 1/\rho$. The general solution is

$$x(\phi) = x_1 + (x_2 - x_1)\,\mathrm{sn}^2\left(\frac{\phi}{2}\sqrt{\rho(x_3 - x_1)} + \delta, k\right) \ ,$$

where $sn(z, k)$ is the Jacobi elliptic function with argument z and modulus $k = \sqrt{(x_2 - x_1)/(x_3 - x_1)}$. The roots can be real ($x_1 \leq x_2 \leq x_3$) or one real x_1 and two complex. If $x_1 < x_2 < x_3$ all real then

$$\frac{d^2x}{d\phi^2} = \frac{\rho}{2}\Big((x - x_2)(x - x_3) + (x - x_1)(x - x_3) + (x - x_1)(x - x_2)\Big) \ ,$$

which is positive for $x = x_1$ and $x = x_3$ and negative for $x = x_2$. Physically, we have an oscillation between x_1 and x_2 or move away from x_3 to infinity. If $x_1 < 0$, a particle coming from infinity get near the mass and move again to infinity. If $x_1 < x_2 = x_3$, calling $r_{in} = 1/x_2$, we have $3/2\rho \leq r_{in} \leq r\rho$. Therefore, there are three solutions: spiral approaching the inner radius; circular orbit at the inner radius; spiral down from the inner radius towards the singularity. Fow $k = 1$, we have $\mathrm{sn}(z, 1) = \tanh(z)$ and

$$x(\phi) = x_1 + (x_2 - x_1)\tanh^2\left(\frac{\phi}{2}\sqrt{\rho(x_2 - x_1)} + \delta\right) \ .$$

If $x_1 = x_2 < x_3$ then the orbit is circular in $r_{out} = 1/x_1$. If $x_2, x_3 \in \mathbb{C}$, then the particle spirals towards a black hole with finite ϕ.

7.5 Weak field phenomena

7.6 Perihelion precession

Consider a circular orbit in radius $r_c = L^2/G_N M$. With $\gamma = \epsilon = 1$, we approximate $r = r_c + \delta r(\tau)$. Expanding the potential, we obtain frequency

$$\omega_r^2 = V''(r_c) = \frac{G_N^4 M^4}{L^6} - \frac{6G_N^6 M^6}{L^8} = \omega_N \Big(1 - \frac{6G_N^2 M^2}{L^2}\Big)$$

where ω_N is the Newtonian frequency. Therefore,

$$\delta\phi_T \simeq \frac{2\pi}{\omega_r} - \frac{2\pi}{\omega_N} \simeq \frac{6\pi G_N^2 M^2}{L^2} \simeq \frac{6\pi G_N M}{r_c} \ .$$

Generalising for elliptic orbits, we have

$$\delta\phi_T \simeq \frac{6\pi G_N M}{a(1-e^2)} \ .$$

7.7 Light deflection

Consider a trajectory of light passing near the Sun with $\epsilon = 0$ and

$$\Big(\frac{r}{\phi}\Big)^2 = \frac{E^2}{L^2} r^4 - r^2 + 2\gamma G_N M r \ .$$

In Newtonian gravity, $\gamma = 0$, light travels in straight line with impact parameter $b = L/E$. Therefore

$$\frac{d\phi}{dr} = \pm \frac{1}{r^2}\Big(\frac{1}{b^2} - \frac{1}{r^2}\Big(1 - \frac{\gamma 2 G_N M}{r}\Big)\Big)^{-1/2}$$

which becomes for $u = 1/r$

$$\frac{d\phi}{du} = \Big(\frac{1}{b^2} - u^2 + 2\gamma G_N M u^3\Big)^{-1/2} \ .$$

For $\gamma = 0$, the solution is

$$r \sin\phi = b \ .$$

For $\gamma = 1$ and $G_N M/b \ll 1$, we define $u = u(1 - G_N M u)$ and expand

$$\frac{d\phi}{dy} \simeq \frac{1 + 2G_N M y}{\sqrt{1/b^2 - y^2}} \ ,$$

which is solved by

$$\phi \simeq \arcsin(by) + \frac{2G_N M}{b} - 2G_N M\sqrt{1/b^2 - y^2} \ .$$

At the minimum $y = 1/b$, we have

$$\phi_b = \frac{\pi}{2} + \frac{2G_N M}{b}$$

and

$$\delta\phi = 2\phi_b - \pi \simeq \frac{4G_N M}{b} \ .$$

Chapter 8

Interior Schwarzschild metric

Together with the exterior metric, Schwarzschild himself provided a solution for the interior of a star. In this chapter, we will study what happens inside a star: interior metric, equation of equilibrium for a spherical symmetric object, called the Tolman-Oppenheimer-Volkov (TOV) equation, Buchdahl limit.

8.1 Computations

It is convenient to write down the static spherical symmetric employing exponentials

$$ds^2 = -\exp(h(r))f(r)dt^2 + f^{-1}(r)dr^2 + r^2 d\Omega^2 \ .$$

In order to find the exterior metric at the radius of the source $r = R_s$, it is reasonable to assume that

$$f(r) = 1 - \frac{2G_N}{r} m(r) \ ,$$

where $m(r)$ it the Misner-Sharp-Hernandez mass function, such that $m(R_s) = M$. In matrix notation, the metric takes the form

$$g = \begin{bmatrix} -f(r)e^{h(r)} & 0 & 0 & 0 \\ 0 & \frac{1}{f(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} \ ,$$

.

The non-vanishing Christoffel symbols are

$$\Gamma^0_{01} = 0.5 \frac{d}{dr} h(r) + \frac{0.5 \frac{d}{dr} f(r)}{f(r)} \ , \quad \Gamma^0_{10} = 0.5 \frac{d}{dr} h(r) + \frac{0.5 \frac{d}{dr} f(r)}{f(r)} \ ,$$

$$\Gamma^1_{00} = 0.5 \left(f(r) \frac{d}{dr} h(r) + \frac{d}{dr} f(r) \right) f(r) e^{h(r)} \ , \quad \Gamma^1_{11} = -\frac{0.5 \frac{d}{dr} f(r)}{f(r)} \ ,$$

$$\Gamma^1_{22} = -1.0rf(r) \ , \quad \Gamma^1_{33} = -1.0rf(r)\sin^2(\theta) \ ,$$

$$\Gamma^2_{12} = \frac{1.0}{r} \ , \quad \Gamma^2_{21} = \frac{1.0}{r} \ ,$$

$$\Gamma^2_{33} = -0.5\sin(2\theta) \ , \quad \Gamma^3_{13} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{23} = \frac{1.0}{\tan(\theta)} \ , \quad \Gamma^3_{31} = \frac{1.0}{r} \ ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} \ , \quad .$$

The non-vanishing Einstein tensor components are

$$G^0{}_0 = \frac{1.0\left(r\frac{d}{dr}f(r) + f(r) - 1\right)}{r^2} \ ,$$

$$G^1{}_1 = \frac{1.0\left(rf(r)\frac{d}{dr}h(r) + r\frac{d}{dr}f(r) + f(r) - 1\right)}{r^2} \ ,$$

$$G^2{}_2 = 0.25f(r)\left(\frac{d}{dr}h(r)\right)^2 + 0.5f(r)\frac{d^2}{dr^2}h(r) + 0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r) + 0.5\frac{d^2}{dr^2}f(r) + \frac{0.5f(r)\frac{d}{dr}h(r)}{r} + \frac{1.0\frac{d}{dr}f}{r}$$

$$G^3{}_3 = 0.25f(r)\left(\frac{d}{dr}h(r)\right)^2 + 0.5f(r)\frac{d^2}{dr^2}h(r) + 0.75\frac{d}{dr}f(r)\frac{d}{dr}h(r) + 0.5\frac{d^2}{dr^2}f(r) + \frac{0.5f(r)\frac{d}{dr}h(r)}{r} + \frac{1.0\frac{d}{dr}f}{r}$$

,

or in terms of m

$$G^0{}_0 = -\frac{2.0G\frac{d}{dr}m(r)}{r^2} \ ,$$

$$G^1{}_1 = \frac{1.0\left(-2.0Gm(r)\frac{d}{dr}h(r) - 2.0G\frac{d}{dr}m(r) + 1.0r\frac{d}{dr}h(r)\right)}{r^2} \ ,$$

$$G^2{}_2 = -\frac{0.5Gm(r)\left(\frac{d}{dr}h(r)\right)^2}{r} - \frac{1.0Gm(r)\frac{d^2}{dr^2}h(r)}{r} - \frac{1.5G\frac{d}{dr}h(r)\frac{d}{dr}m(r)}{r} - \frac{1.0G\frac{d^2}{dr^2}m(r)}{r} + \frac{0.5Gm(r)\frac{d}{dr}h(r)}{r^2}$$

$$G^3{}_3 = -\frac{0.5Gm(r)\left(\frac{d}{dr}h(r)\right)^2}{r} - \frac{1.0Gm(r)\frac{d^2}{dr^2}h(r)}{r} - \frac{1.5G\frac{d}{dr}h(r)\frac{d}{dr}m(r)}{r} - \frac{1.0G\frac{d^2}{dr^2}m(r)}{r} + \frac{0.5Gm(r)\frac{d}{dr}h(r)}{r^2}$$

.

8.2 Interior metric

Suppose we are dealing with a perfect fluid with energy density $\rho(r)$ and pressure $p(r)$. The Einstein field equations reads as

$$G^0{}_0 = -\frac{2G_N m'}{r^2} = -8\pi G_N \rho \ ,$$

$$G^1{}_1 = \frac{fh'}{r} + \frac{f'}{r} + \frac{f}{r} - \frac{1}{r^2} = 8\pi G_N p \ .$$

The first Einstein field equation is solved by

$$m'(r) = 4\pi\rho(r)r^2 \ , \quad m(r) = 4\pi \int_0^r dr \ \rho(r)r^2 \ .$$

Another assumption that we make is homogeneity, i.e. the energy density is constant inside the star for $r \in [0, R_s]$

$$\rho(r) = \rho_0 = \frac{3M}{4\pi R_s^3} \ .$$

Therefore, the mass function becomes

$$m(r) = 4\pi \int_0^r dr \ \rho_0 r^2 = 4\pi \frac{3M}{4\pi R_s^3} \int_0^r dr \ r^2 = \frac{Mr^3}{R_s^3}$$

and

$$f(r) = 1 - \frac{2G_N}{r} m(r) = 1 - \frac{2G_N r^2}{R_s^3} \ , \quad f'(r) = \frac{4G_N r}{R_s^3} \ .$$

Now, notice that if we subtract the second to the first, we obtain

$$\frac{fh'}{r} = 8\pi G_N (p + \rho) \ , \quad h' = 8\pi G_N r f^{-1}(p + \rho) \ .$$

However, it is difficult to solve this equation since also the pressure is unknown. In the next section, we find a way to determine that by means of the continuity equation.

8.3 TOV equation

Using the Christoffel symbols calculated before, the Bianchi identity reads as

$$\begin{aligned} 0 = \nabla_\mu T^\mu{}_1 &= \partial_\mu T^\mu{}_1 + \Gamma^\mu_{\nu\mu} T^\nu{}_1 - \Gamma^\nu_{1\mu} T^\mu{}_\nu \\ &= p' + \Gamma^0_{01} p + \Gamma^1_{11} p + \Gamma^2_{21} p + \Gamma^3_{31} p + \Gamma^0_{01} \rho - \Gamma^1_{11} p - \Gamma^2_{21} p - \Gamma^3_{31} p \\ &= p' + \Gamma^0_{01}(\rho + p) = p' + \frac{1}{2}\Big(h' + \frac{f'}{f}\Big)(\rho + p) \ , \end{aligned}$$

which implies that, using the expressions

$$f = 1 - \frac{2G_N}{r} m \ , \quad h' = 8\pi G_N r f^{-1}(p+\rho) \ ,$$

$$f' = -\frac{2G_N}{r} m' + \frac{2G_N}{r^2} m = -\frac{2G_N}{r} 4\pi\rho r^2 + \frac{2G_N}{r^2} m = -8\pi G_N r\rho + \frac{2G_N}{r^2} m \ ,$$

that

$$\begin{aligned} p' &= -\frac{1}{2}\Big(8\pi G_N r f^{-1}(p+\rho) + \frac{1}{f}(-8\pi G_N r\rho + \frac{2G_N}{r^2} m)\Big)(\rho+p) \\ &= -f^{-1}(4\pi G_N r p + \frac{G_N}{r^2} m)(\rho+p) \\ &= -\frac{G_N}{r^2}\Big(1 - \frac{2G_N m}{r}\Big)^{-1}(m + 4\pi r^3 p)(\rho+p) \ , \end{aligned}$$

which is the so-called TOV equation. Notice that in the non-relativistic limit, it becomes the Newtonian hydrostatic equilibrium equation

$$p' \simeq \rho(r)\frac{G_N m}{r^2} = -\rho(r) V_N'(r) \ .$$

In our homogeneous case, the pressure becomes

$$\begin{aligned} p' &= -\frac{G_N}{r^2}\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{-1}(\frac{M r^3}{R_s^3} + 4\pi r^3 p)(\rho_0 + p) \\ &= -4\pi G_N r\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{-1}(\frac{\rho}{3} + p)(\rho_0 + p) \\ &= -\frac{4}{3}\pi G_N r\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{-1}(\rho + 3p)(\rho_0 + p) \ , \end{aligned}$$

$$-\frac{4}{3}\pi G_N r\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{-1} = \frac{p'}{(\rho+3p)(\rho_0+p)} \ ,$$

$$\frac{1}{4\rho_0}\frac{d}{dr}\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big) = \frac{1}{2\rho_0}\frac{d}{dr}\ln\frac{\rho_0+3p}{\rho+p} \ ,$$

$$\frac{1}{4\rho_0}\frac{d}{dr}\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big) = \frac{1}{2\rho_0}\frac{d}{dr}\ln\frac{\rho_0+3p}{\rho_0+p} \ ,$$

$$\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2} = C\frac{\rho_0+3p}{\rho_0+p} \ ,$$

$$\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2}\rho_0 + \Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2} p = C\rho_0 + 3Cp \ ,$$

$$(\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2} - 3C)p = (C - \Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2})\rho_0 \ .$$

$$p = \rho_0(C - \Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2})(\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2} - 3C)^{-1} \ .$$

Finally, imposing initial conditions $p(R_s) = 0$, we find

$$(C - \Big(1 - \frac{2G_N M}{R_s}\Big)^{1/2})(\Big(1 - \frac{2G_N M}{R_s}\Big)^{1/2} - 3C)^{-1} = 0 \ ,$$

$$C = \Big(1 - \frac{2G_N M}{R_s}\Big)^{1/2} \ .$$

Hence, we obtain

$$\begin{aligned} p(r) &= \rho_0(\Big(1 - \frac{2G_N M}{R_s}\Big)^{1/2} - \Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2})(\Big(1 - \frac{2G_N M r^2}{R_s^3}\Big)^{1/2} - 3\Big(1 - \frac{2G_N M}{R_s}\Big)^{1/2})^{-1} \\ &= \rho_0 \frac{(1 - 2G_N M/R_s)^{1/2} - (1 - 2G_N M r^2/R_s^3)^{1/2}}{(1 - 2G_N M r^2/R_s^3)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}} \\ &. \end{aligned}$$

Now, we can explicitly solve the Einstein field equations. Using the continuity equation

$$0 = p' + \frac{1}{2}\Big(h' + \frac{f'}{f}\Big)(\rho + p)$$

we can extract a derivative

$$(h + \ln f)' = -\frac{2p'}{p + \rho} \ ,$$

and integrate

$$\int d(h + \ln f) = -\int \frac{2dp}{p + \rho} \ , \quad h + \ln f = -2\ln(p + \rho) \ ,$$

$$\exp(h/2) f^{1/2} = \frac{C}{p + \rho} \ .$$

Now, we set C by suitable initial conditions $h(R_s) = 0$, and we notice

$$\begin{aligned} p + \rho_0 &= \rho_0 \frac{(1 - 2G_N M/R_s)^{1/2} - (1 - 2G_N M r^2/R_s^3)^{1/2}}{(1 - 2G_N M r^2/R_s^3)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}} + \rho_0 \\ &= \rho_0 \frac{(1 - 2G_N M/R_s)^{1/2} - (1 - 2G_N M r^2/R_s^3)^{1/2} + (1 - 2G_N M r^2/R_s^3)^{1/2} - 3(1 - 2G_N M/R_s)^{1}}{(1 - 2G_N M r^2/R_s^3)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}} \\ &= \rho_0 \frac{(1 - 2G_N M/R_s)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}}{(1 - 2G_N M r^2/R_s^3)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}} \\ &= -\rho_0 \frac{2(1 - 2G_N M/R_s)^{1/2}}{(1 - 2G_N M r^2/R_s^3)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}} \ . \end{aligned}$$

Hence, we find

$$\exp(h/2) = (1 - 2G_N M r^2/R_s^3)^{-1/2} \frac{C}{\rho_0} \frac{(1 - 2G_N M r^2/R_s^3)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}}{2(1 - 2G_N M/R_s)^{1/2}} \ ,$$

where C can be determined by

$$1 = (1 - 2G_N M/R_s)^{-1/2} \frac{C}{\rho_0} \frac{(1 - 2G_N M/R_s)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}}{2(1 - 2G_N M/R_s)^{1/2}} \ ,$$

$$C = \rho_0 \frac{2(1 - 2G_N M/R_s)}{(1 - 2G_N M/R_s)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}} \ ,$$

Finally,

$$\exp(h/2) = (1 - 2G_N M r^2/R_s^3)^{-1/2} \frac{(1 - 2G_N M/R_s)^{1/2}}{(1 - 2G_N M/R_s)^{1/2} - 3(1 - 2G_N M/R_s)^{1/2}} \frac{(1 - 2G_N M r^2/R_s^3)^{1/2} -}{1}$$

8.4 Buchdahl limit

The pressure is maximal at $r = 0$

$$p(0) = \rho_0 \frac{\sqrt{1 - 2G_N M/R_s} - 1}{1 - 3\sqrt{1 - 2G_N M/R_s}}$$

and diverges for

$$R_S = \frac{9}{4} G_N M = \frac{9}{8} R_H \ .$$

Physically, it means that for $R < R_s$, the star is not in equilibrium.

Part IV

Cosmology

Chapter 9

Friedmann-Robertson-Walker

Cosmology is the study of the Universe as a whole. Since the only non-negligible interaction for the study of the large-scale structure of the Universe is gravity, we can study the Cosmos by means of the general theory of relativity. In this chapter, after some reasonable assumptions of isotropy and homogeneity for the cosmic fluid, we will derive a suitable metric to mathematically describe the Universe, the so-called Friedmann-Robertson-Walker or FRW metric, and find the corresponding Einstein field equations, the Friedmann equations. Besides, we will delve into the solutions of these equations in the case of open, flat or close Universe.

9.1 Principles

Our starting point is two principles upon which modern cosmology is based.

Principle 9.1 (Copernican)
There is no preferred observer in the Universe.

Principle 9.2 (Cosmological)
The Universe is homogeneous and isotropic.

Actually, the former is implied by homogeneity of the latter: if there are no preferred observers, the Universe looks the same from any point, thus they can all be considered equivalent. Every assumption should be experimentally or observationally justified. Of course, isotropy could be tested by looking at the night sky, but what about homogeneity? Actually, even the night sky looks everything but isotropic, but we can restrict our attention to little sampled areas and checking the average distribution of matter in all directions. In addition, we have to take into account the fact that the image that we see at a given instant of time is in reality is the past light-cone. This implies that the Cosmological principle must be reformulated.

Principle 9.3 (Cosmological II)
There exists a time t such that the Universe is homogeneous and isotropic on each time slice Σ_t of it.

Therefore, the Cosmological principle need to be considered as a working hypothesis and to be verified a posteriori.

9.2 Symmetries

The metric has 10 independent components, but it can be much simplified by studying the symmetries of the system. Isotropy entails the existence of three rotational space-like Killing vectors K_a with $a = 1, 2, 3$ such that they satisfy the Lie algebra $[K_a, K_b] = 0$; homogeneity entails the existence of three translational space-like Killing vectors K_i with $i = 1, 2, 3$ such that they satisfy the Lie algebra $[K_i, K_j] = \epsilon_{ijk} K_k$. We are in the so-called maximal symmetry and there are no more isometries in space. In fact, it can be proved that isotropy with respect to an arbitrary point is equivalent to homogeneity. Concerning time, there is no Killing vector associated to it. Accordingly, the metric depends only on a function of (coordinate) time $a = a(t)$, called the cosmic scale factor, to be determined by the Einstein field equations. Adopting as comoving coordinates (t, r, θ, ϕ), the metric explicitly looks like

$$ds^2 = -dt^2 + a^2(t)\Big(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2\Big) \ .$$

This is known as the Friedmann-Robertson-Walker or FRW metric. Notice that there is a factor k, called the curvature scalar, that in principle could take any real value. However, it is always possible to rescale the r coordinate in order to have $r = \tilde{r}/|k|^{1/2}$, so that

$$\begin{aligned} ds^2 &= -dt^2 + a^2(t)\Big(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2\Big) \\ &= -dt^2 + a^2(t)\Big(\frac{d\tilde{r}^2/|k|}{1 - \tilde{r}^2 k/|k|} + \frac{\tilde{r}^2}{|k|} d\Omega^2\Big) \\ &= -dt^2 + \frac{a^2(t)}{|k|}\Big(\frac{dr^2}{1 - \tilde{r}^2 k/|k|} + r^2 d\Omega^2\Big) \\ &= -dt^2 + \tilde{a}^2(t)\Big(\frac{dr^2}{1 - \tilde{r}^2 \tilde{k}} + r^2 d\Omega^2\Big) \ , \end{aligned}$$

where we have rescaled a by $\tilde{a} = a/|k|$ and $\tilde{k} = k/|k|$. In this way, the only values that k can take are $0, \pm 1$. To understand the physical meaning of these values, let us make a change of variable of r into ψ such that

$$ds^2 = -dt^2 + a^2(t)(d\psi^2 + f_k(\psi)^2 d\Omega^2) \ ,$$

where

$$f_k(\psi) = \begin{cases} \psi & k = 0 \\ \sin\psi & k = 1 \\ \sinh\psi & k = -1 \end{cases} .$$

Depending on the sign of the curvature scalar k, we have

1. if $k = 0$, a flat universe with metric

$$ds^2 = -dt^2 + a^2(t)(dr^2 + r^2 d\Omega^2) = -dt^2 + a^2(t)(d\psi^2 + \psi^2 d\Omega^2)$$

   and hypersurface at constant time Σ_t to be flat;

2. if $k = 1$, an open universe with metric

$$ds^2 = -dt^2 + a^2(t)\Big(\frac{dr^2}{1-r^2} + r^2 d\Omega^2\Big) = -dt^2 + a^2(t)(d\psi^2 + \sin^2\psi d\Omega^2)$$

   and hypersurface at constant time Σ_t to be a 3-dimensional sphere;

3. if $k = -1$, a closed universe with metric

$$ds^2 = -dt^2 + a^2(t)\Big(\frac{dr^2}{1+r^2} + r^2 d\Omega^2\Big) = -dt^2 + a^2(t)(d\psi^2 + \sinh^2\psi d\Omega^2)$$

   and hypersurface at constant time Σ_t to be a 3-dimensional hyperboloid.

Now, we can express which is the meaning of the coordinates. The areal radius, i.e. the radius such that the area is given by $A = 4\pi r_a^2$, is given by

$$r_a = a(t)r \ ,$$

whereas the proper distance is

$$dR = \sqrt{g_{00}}dr = \frac{a(t)}{\sqrt{1-kr^2}}dr \ .$$

Notice that both distances depend on time. a gives the size of the Universe: for $k = 1$, it is the radius of the Universe-sphere; for $k = 0$, we can substitute the flat Universe with a torus of size a; for $k = -1$, even though such interpretation is not possible, it gives the size of the curvature of the Universe-hyperboloid. Therefore, as observations suggest, the Universe is expanding through $a(t)$.

9.3 Computations

In matrix notation, the metric takes the form

$$g = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -\frac{a^2(t)}{kr^2-1} & 0 & 0 \\ 0 & 0 & r^2a^2(t) & 0 \\ 0 & 0 & 0 & r^2a^2(t)\sin^2(\theta) \end{bmatrix} ,$$

.

The non-vanishing Christoffel symbols are

$$\Gamma^0_{11} = -\frac{1.0a(t)\frac{d}{dt}a(t)}{kr^2-1} , \quad \Gamma^0_{22} = 1.0r^2a(t)\frac{d}{dt}a(t) ,$$

$$\Gamma^0_{33} = 1.0r^2a(t)\sin^2(\theta)\frac{d}{dt}a(t) , \quad \Gamma^1_{01} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} ,$$

$$\Gamma^1_{10} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} , \quad \Gamma^1_{11} = -\frac{1.0kr}{kr^2-1} ,$$

$$\Gamma^1_{22} = 1.0r\left(kr^2-1\right) , \quad \Gamma^1_{33} = 1.0r\left(kr^2-1\right)\sin^2(\theta) ,$$

$$\Gamma^2_{02} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} , \quad \Gamma^2_{12} = \frac{1.0}{r} ,$$

$$\Gamma^2_{20} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} , \quad \Gamma^2_{21} = \frac{1.0}{r} ,$$

$$\Gamma^2_{33} = -0.5\sin(2\theta) , \quad \Gamma^3_{03} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} ,$$

$$\Gamma^3_{13} = \frac{1.0}{r} , \quad \Gamma^3_{23} = \frac{1.0}{\tan(\theta)} ,$$

$$\Gamma^3_{30} = \frac{1.0\frac{d}{dt}a(t)}{a(t)} , \quad \Gamma^3_{31} = \frac{1.0}{r} ,$$

$$\Gamma^3_{32} = \frac{1.0}{\tan(\theta)} , \quad .$$

The non-vanishing Einstein tensor components are

$$G^0{}_0 = \frac{3.0\left(-k-\left(\frac{d}{dt}a(t)\right)^2\right)}{a^2(t)} ,$$

$$G^1{}_1 = \frac{-1.0k-2.0a(t)\frac{d^2}{dt^2}a(t)-1.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} ,$$

$$G^2{}_2 = \frac{-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$
$$G^3{}_3 = \frac{-1.0k - 2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \ ,$$
$$.$$

9.4 Cosmic fluid

Having identified the geometrical part of the system, i.e. the metric, it is time to study the matter content. Our assumption here is that the Universe is a perfect fluid, whose energy-momentum tensor looks like

$$T_{\mu\nu}(t) = (p(t) + \rho(t))u_\mu u_\nu + p(t)g_{\mu\nu} \ , \quad T^\mu{}_\nu(t) = \mathrm{diag}(-\rho(t), p(t), p(t), p(t)) \ ,$$

where u_μ is the comoving 4-velocity, i.e. the fluid is at rest with it, $\rho(t)$ is the energy density and $p(t)$ is the pressure. The 0-th component of the continuity equation for the energy-momentum tensor is the energy conservation

$$\dot\rho + 3H(p + \rho) = 0 \ ,$$

where $H(t) = \dot a(t)/a(t)$ is the Hubble constant.

Proof. In fact, the 0-th component is given by

$$\begin{aligned} 0 = \nabla_\mu T^\mu{}_0 &= \partial_\mu T^\mu{}_0 + \Gamma^\mu_{\nu\mu} T^\nu{}_0 - \Gamma^\nu_{0\mu} T^\mu{}_\nu \\ &= \partial_0 T^0{}_0 + \Gamma^1_{01} T^0{}_0 + \Gamma^2_{02} T^0{}_0 + \Gamma^3_{03} T^0{}_0 - \Gamma^1_{01} T^1{}_1 - \Gamma^2_{02} T^2{}_2 - \Gamma^3_{03} T^3{}_3 \ . \end{aligned}$$

Now, we use the non-vanishing Christoffel symbols

$$\Gamma^1_{01} = \Gamma^2_{02} = \Gamma^3_{03} = \frac{\dot a}{a}$$

to obtain

$$0 = -\dot\rho - 3\frac{\dot a}{a}\rho - 3\frac{\dot a}{a}p = -\dot\rho - 3H\rho - 3Hp = -\dot\rho - 3H(\rho + p) \ .$$

q.e.d.

In order to prove that it is indeed an energy conservation law, notice that we can define a comoving volume $V(t) \sim a^3(t)$ such that the total comoving energy is $E(t) = \rho(t)V(t)$. In fact

$$\begin{aligned} \frac{dE}{dt} &= \frac{d}{dt}(\rho V) = \frac{d}{dt}(\rho a^3) = \dot\rho a^3 + \rho\frac{d}{dt}a^3 = -3\frac{\dot a}{a}(\rho + p)a^3 + 3\rho\dot a a^2 \\ &= -3\frac{\dot a}{a}(\rho + p)a^3 + 3\rho\frac{\dot a}{a}a^3 = -3\frac{\dot a}{a}pa^3 = -p\frac{d}{dt}a^3 = -p\frac{dV}{dt} \ , \end{aligned}$$

which is the thermodynamics relation $dE = -pdV$.

In addition, we need to specify an equation of state, which relate pressure and energy density $p = p(\rho)$ for a barytropic fluid. The most simple relation we can have is linear

$$p = \omega\rho \ ,$$

for some constant parameter ω to be determined. The solution of the energy conservation is

$$\rho \propto a^{-3(1+\omega)} \ .$$

Proof. In fact,

$$0 = \dot{\rho} + 3H(\rho + p) = \dot{\rho} + 3H\rho(1+\omega) \ ,$$

$$\frac{\dot{\rho}}{\rho} = -3(1+\omega)\frac{\dot{a}}{a} \ , \quad \frac{d\rho}{\rho} = -3(1+\omega)\frac{da}{a} \ ,$$

$$\int \frac{d\rho}{\rho} = -3(1+\omega)\int \frac{da}{a} \ , \quad \ln\rho \propto -3(1+\omega)\ln a \ ,$$

$$\ln\rho \propto \ln a^{-3(1+\omega)} \ , \quad \rho \propto a^{-3(1+\omega)} \ .$$

q.e.d.

The cosmic fluid can be decomposed into two part: dust, i.e. non-relativistic pressure-less particles, and radiation, i.e. highly-relativistic particles

1. dust, i.e. non-relativistic pressure-less particles, with $\omega = 0$ and $\rho \propto a^{-3}$;
2. radiation, i.e. non-relativistic pressure-less particles, with $\omega = 1/3$ and $\rho \propto a^{-4}$;
3. dark energy, compatible with the acceleration of the Universe, with $\omega = -1$ and $\rho \propto 1$.

The value of dust is justified by the fact that $p = 0$ implies $\omega = 0$. The value of radiation is justified by the fact that electromagnetic radiation is described by the traceless Maxwell tensor, which implies that the energy-momentum tensor must be traceless $\mathrm{tr}\, T = -\rho + 3p = 0$ so that $p = \rho/3$. Notice that they both agree with special relativistic considerations: for dust $E \sim m_0$ and $V \sim a^3$, whereas for radiation with red-shift $E \propto a^{-1}$. Furthermore, the dark energy contribution can be written in terms of the famous cosmological constant Λ

$$\rho = -p = \frac{\Lambda}{8\pi G_N} \ .$$

Proof. In fact, using the Einstein field equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ , \quad G_{\mu\nu} = 8\pi G_N T_{\mu\nu} - \Lambda g_{\mu\nu} \ ,$$

$$G_{\mu\nu} = 8\pi G_N (T_{\mu\nu} - \frac{\Lambda g_{\mu\nu}}{8\pi G_N}) = 8\pi G_N (T_{\mu\nu} - T^{\Lambda}_{\mu\nu}) \ ,$$

hence, we have

$$T^{\Lambda}_{00} = \rho = \frac{\Lambda}{8\pi G_N} = -p \ .$$

q.e.d.

9.5 Friedmann equations

It is time to calculate the Einstein field equations for the FRW metric, which are called the Friedmann equations

$$3\Big(\Big(\frac{\dot{a}}{a}\Big)^2 + \frac{k}{a^2}\Big) = 8\pi G_N \rho \ ,$$

$$3\frac{\ddot{a}}{a} = -4\pi G_N (\rho + 3p) \ .$$

Notice that the curvature scalar k appears only in the form k/a^2; in this way a rescaling is well-compensated. The first one is only a constraint to select initial conditions, while it is the second equation that governs the time evolution.

Proof. Because of isotropy, there are only 2 independent equations. For the 00-th components of the Einstein tensor

$$G^0{}_0 = g^{00} G_{00} = -\frac{3}{a^2}(k + \dot{a}^2) \ ,$$

we obtain

$$G^0{}_0 = 8\pi G_N T^0{}_0 \ , \quad -\frac{3}{a^2}(k + \dot{a}^2) = -8\pi G_N \rho \ ,$$

$$3\Big(\Big(\frac{\dot{a}}{a}\Big)^2 + \frac{k}{a^2}\Big) = 8\pi G_N \rho \ .$$

For the 11-th components of the Einstein tensor

$$G^1{}_1 = g^{11} G_{11} = -\frac{1}{a^2}(k + 2a\ddot{a} + \dot{a}^2) \ ,$$

we obtain

$$G^1 = 8\pi G_N T^1{}_1 \ , \quad -\frac{1}{a^2}(k + 2a\ddot{a} + \dot{a}^2) = 8\pi G_N p \ .$$

Now, we use the first equation

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8}{3}\pi G_N \rho$$

to simplify the second one

$$-\frac{1}{a^2}(k + 2a\ddot{a} + \dot{a}^2) = -\left(\frac{\dot{a}}{a}\right)^2 - \frac{k}{a^2} - 2\frac{\ddot{a}}{a} = -\frac{8}{3}\pi G_N \rho - 2\frac{\ddot{a}}{a} \ .$$

Hence, we find

$$-\frac{8}{3}\pi G_N \rho - 2\frac{\ddot{a}}{a} = 8\pi G_N p \ , \quad 6\frac{\ddot{a}}{a} = -24\pi G_N p - 8\pi G_N \rho \ ,$$

$$3\frac{\ddot{a}}{a} = -4\pi G_N(\rho + 3p) \ .$$

q.e.d.

Because of the Bianchi identity, the energy conservation law and the Friedmann equations are not independent.

Proof. In fact, deriving the first one

$$\frac{d}{dt}\frac{\dot{a}^2}{a^2} + \frac{d}{dt}\frac{k}{a^2} = -2\frac{\dot{a}^2}{a^3} + 2\frac{\ddot{a}}{a^2} - 2\frac{k}{a^3} = \frac{8}{3}\pi G_N \dot{\rho} \ ,$$

then using the energy conservation law, we find

$$-2\frac{\dot{a}^2}{a^3} + 2\frac{\ddot{a}}{a^2} - 2\frac{k}{a^3} = -8\pi G_N(\rho + p)\frac{\dot{a}}{a} \ , \quad -\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{k}{a^2} = 4\pi G_N(\rho + p)\dot{a} \ ,$$

Finally, using the first one, we obtain

$$-\frac{\ddot{a}}{a} + \frac{8}{3}\pi G_N = 4\pi G_N(\rho + p)\dot{a} \ , \quad 3\frac{\ddot{a}}{a} = -4\pi G_N(\rho + 3p) \ .$$

Similarly, it can be shown that they are all related. q.e.d.

It is useful to express the Friedmann equations in terms of the density parameter and critical density

$$\Omega = \frac{8\pi G_N \rho}{3H^2} = \frac{\rho}{\rho_{crit}} \ , \quad \rho_{crit} = \frac{3H^2}{8\pi G_N} \ .$$

Therefore, the first Friedmann equation becomes

$$\Omega - 1 = \frac{k}{H^2 a^2} \ .$$

Proof. Expressing the first equation in terms of H, we find

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = H^2 + \frac{k}{a^2} = \frac{8}{3}\pi G_N \rho \ , \frac{k}{a^2} = \frac{8}{3}\pi G_N \rho - H^2 \ ,$$

$$\frac{k}{a^2 H^2} = \frac{8\pi G_N \rho}{3H^2} - 1 = \Omega - 1 \ .$$

q.e.d.

As we said before, the sign of the curvature scalar k can be used to determine which is the shape of the Universe. However, by means of this equation, we can equivalently use the density parameter Ω or the density ρ in the following way

1. open universe if $k = -1$ or $\rho < \rho_{crit}$ or $\Omega < 1$,
2. flat universe if $k = 0$ or $\rho = \rho_{crit}$ or $\Omega = 1$,
3. closed universe if $k = 1$ or $\rho > \rho_{crit}$ or $\Omega > 1$.

Now, we are able to solve the Friedmann equations for the simple flat case:

1. flat matter-dominated, for which

$$k = 0 \ , \quad \rho \sim a^{-3} \quad \Rightarrow \quad a \sim t^{2/3} \ ;$$

2. flat radiation-dominated, for which

$$k = 0 \ , \quad \rho \sim a^{-4} \quad \Rightarrow \quad a \sim t^{1/2} \ ;$$

3. flat dark energy-dominated, for which

$$k = 0 \ , \quad \rho \sim \Lambda \quad \Rightarrow \quad a \sim \exp(H_0 t) \ .$$

Proof. For the flat matter-dominated, i.e. $k = 0$ or $\Omega = 1$ and $\rho \sim a^{-3}$, we have

$$1 = \Omega \sim \frac{\rho}{H^2} = \frac{1}{a^3}\frac{a^2}{\dot{a}^2} = \frac{1}{a\dot{a}^2} \ , \quad a\dot{a}^2 = 1 \ ,$$

$$a^{1/2}\dot{a} \sim 1 \ , \quad a^{1/2} da \sim dt \ , \quad a^{3/2} \sim t \ , \quad a \sim t^{2/3} \ .$$

For the flat radiation-dominated, i.e. $k = 0$ or $\Omega = 1$ and $\rho \sim a^{-4}$, we have

$$1 = \Omega \sim \frac{\rho}{H^2} = \frac{1}{a^4}\frac{a^2}{\dot{a}^2} = \frac{1}{a^2\dot{a}^2} \ , \quad a^2\dot{a}^2 = 1 \ ,$$

$$a\dot{a} \sim 1 \ , \quad a da \sim dt \ , \quad a^2 \sim t \ , \quad a \sim t^{1/2} \ .$$

For the flat dark energy-dominated, i.e. $k = 0$ or $\Omega = 1$ and $\rho \sim \Lambda$, we have

$$1 = \Omega \sim \frac{\rho}{3H^2} = \frac{\Lambda}{3}\frac{a^2}{\dot{a}^2} \ , \quad \dot{a}^2 = \frac{\Lambda}{3}a^2 \ ,$$

$$\dot{a} \sim (\Lambda/3)^{1/2}a \ , \quad \frac{da}{a} \sim (\Lambda/3)^{1/2}dt \ , \quad a \sim \exp((\Lambda/3)^{1/2}t) = \exp(H_0 t) \ .$$

q.e.d.

We conclude the chapter with a comparison between the Einstein and the Newton cosmology. Consider a particle of mass m on the surface of radius $R = ra$. The Newtonian second law reads as

$$\ddot{R} = r\ddot{a} = -V_N = -\frac{G_N M}{R^2} = -\frac{G_N \rho}{R^2}V = -\frac{G_N \rho}{R^2}\frac{4}{3}\pi R^3 = -\frac{4}{3}\pi G_N \rho R = -\frac{4}{3}\pi G_N \rho r a \ ,$$

$$3\frac{\ddot{a}}{a} = -4\pi G_N \rho \ ,$$

which shows that the Newtonian limit can be reached only if $p = 0$.

Chapter 10

Expansion of the universe

Hubble's law states that galaxies are moving away from us at a velocity that is proportional to their distance. It is important because it gives a direct observation of the Hubble constant. In this chapter, we will study how to derive this law from observational quantities: luminosity and red-shift. In the final section, we will briefly review what is called the standard model of cosmology: cosmic background radiation, inflation and Λ-CDM model.

10.1 Cosmological red-shift

Place the origin at $r = 0$. Consider a comoving source at r_1 and a comoving receiver at r_2. Two emitted radially (θ and ϕ constant) electromagnetic waves at times t_1 and $t_1 + \delta t_1$ will be received at times t_2 and $t_2 + \delta t_2$. We want to calculate the difference in frequency between them. Since light has zero interval, we have

$$0 = ds^2 = -dt^2 + \frac{a^2(t)}{1 - kr^2} dr^2 \ , \quad dt = \frac{a(t)}{\sqrt{1 - kr^2}} dr \ .$$

For the first signal, we have

$$\int_{r_1}^{r_2} \frac{dr}{\sqrt{1 - kr^2}} = \int_{t_1}^{t_2} \frac{dt}{a(t)} \ ,$$

whereas, for the second signal, we have

$$\int_{r_1}^{r_2} \frac{dr}{\sqrt{1 - kr^2}} = \int_{t_1+\delta t_1}^{t_2+\delta t_2} \frac{dt}{a(t)} \ .$$

Equating them two, we find

$$\int_{t_1}^{t_2} \frac{dt}{a(t)} = \int_{t_1+\delta t_1}^{t_2+\delta t_2} \frac{dt}{a(t)} \ ,$$

$$\int_{t_1}^{t_1+\delta t_1} \frac{dt}{a(t)} + \cancel{\int_{t_1+\delta t_1}^{t_2} \frac{dt}{a(t)}} = \cancel{\int_{t_1+\delta t_1}^{t_2} \frac{dt}{a(t)}} + \int_{t_2}^{t_2+\delta t_2} \frac{dt}{a(t)} \ ,$$

$$\int_{t_1}^{t_1+\delta t_1} \frac{dt}{a(t)} = \int_{t_2}^{t_2+\delta t_2} \frac{dt}{a(t)} \ .$$

Making the reasonable assumption that δt_1 and δt_2 are so small that the cosmic scale factor a does not change significantly, then we obtain

$$\frac{\delta t_1}{a(t_1)} = \frac{\delta t_2}{a(t_2)} \ , \quad \frac{\delta t_1}{\delta t_2} = \frac{a(t_1)}{a(t_2)} \ .$$

It is useful to rewrite in terms of frequencies

$$\frac{\omega(t_1)}{\omega(t_2)} = \frac{2\pi}{\delta t_1}\frac{\delta t_2}{2\pi} = \frac{\delta t_2}{\delta t_1} = \frac{a(t_2)}{a(t_1)} \ ,$$

or in terms of the red-shift z, defined as

$$z = \frac{\lambda_2 - \lambda_1}{\lambda_1} = \frac{a(t_2)}{a(t_1)} - 1 \ ,$$

where λ_2 is the observed wavelength and λ_1 is emitted observed. This red-shift is from different nature than Doppler effect, since it is caused by the expansion of spacetime. If the Universe is expanding, i.e. $a(t_2) > a(t_1)$, the red-shift is positive, otherwise if the Universe is contracting, i.e. $a(t_2) < a(t_1)$, the red-shift is negative, which is called blue-shift. It is symmetric between observer and receiver, unlike the gravitational red-shift. Furthermore, it is important highlight that $a(t)$ can be rescaled, therefore is not physical, but in our formula it is the ratio that compares. It depends only on time of emission and arrival (it is not cumulative); it is due a different reference frame used by them two.

10.2 Luminosity-distance relation

In astronomy, the only physical quantity we can directly measure is the apparent luminosity of a stellar object. However, we can measure also distance using the luminosity-distance relation.

Consider first the Minkowski metric. Since energy is conserved during propagation, the flux of energy through a surface of radius R is time-independent and depends only on radius

$$F = \frac{L}{A} = \frac{L_0}{4\pi R^2} \ .$$

Therefore, the luminosity-distance relation is

$$d_L^2 = R^2 = \frac{L_0}{4\pi F} \ .$$

Consider now the FRW metric. In the previous section, we have seen that photons are red-shifted by a factor $(1+z)$. In particular, the observer receives a signal that is delayed by a factor $(1+z)^2$ and the luminosity transforms as

$$L \simeq \frac{L_0}{(1+z)^2} \ ,$$

hence, the flux becomes

$$F = \frac{L}{A} = \frac{L}{4\pi R^2} = \frac{L}{4\pi (a_0 r)^2} \simeq \frac{L_0}{4\pi (a_0 r)^2 (1+z)^2} = \frac{L_0}{4\pi d_L^2} \ ,$$

where the luminosity-distance relation is given by

$$d_L = a_0 r (1+z) \simeq \sqrt{\frac{L_0}{F}} \ . \tag{10.1}$$

Notice that for $z > 0$, we have darker objects. Besides, in order to measure the red-shift, we need the emitted original frequency. In astronomy, this can be achieved by using standard candles: characteristic behaviours, like period of oscillation of luminosity.

10.3 Hubble's law

After we have introduced these two observable quantities, we are able to extract the Hubble law. The first assumption is that we are working with small red-shift so that we can work in power series in z. Therefore, we expand the cosmic scale factor in time of emission $t = t_1$ around t_0

$$a(t) = a_0 + \dot{a}_0 (t - t_0) + \frac{\ddot{a}_0}{2}(t-t_0)^2 + O((t-t_0)^3) \ .$$

We introduce the deceleration parameter

$$q = -\frac{a\ddot{a}}{\dot{a}^2}$$

and rewrite the expansion in terms of q_0 and H_0

$$\begin{aligned} a(t) &= a_0 \Big(1 + \frac{\dot{a}_0}{a_0}(t-t_0) + \frac{1}{2}\frac{\dot{a}_0^2}{a_0^2}\frac{\ddot{a}_0 a_0}{\dot{a}_0^2}(t-t_0)^2 \Big) + O((t-t_0)^3) \\ &= a_0 \Big(1 + H_0 (t-t_0) - \frac{1}{2} H_0^2 q_0 (t-t_0)^2 \Big) + O((t-t_0)^3) \ . \end{aligned}$$

Therefore, the red-shift can be expanded as

$$\begin{aligned} z &= \frac{a_0}{a} - 1 \simeq \Big(1 + H_0(t-t_0) - \frac{1}{2} H_0^2 q_0 (t-t_0)^2 \Big)^{-1} - 1 + O((t-t_0)^3) \\ &\simeq 1 - H_0(t-t_0) + \frac{1}{2} H_0^2 q_0 (t-t_0)^2 + H_0^2 (t-t_0)^2 - 1 + O((t-t_0)^3) \\ &= -H_0(t-t_0) + H_0^2 \Big(\frac{q_0}{2} + 1 \Big)(t-t_0)^2 + O((t-t_0)^3) \ , \end{aligned}$$

where we have used the expansion

$$\frac{1}{1+x} = 1 - x + x^2 + O(x^3) \ .$$

Now, we need to invert this relation in order to find $t_0 - t$ in terms of z. Through an ansatz

$$t_0 - t = Az + Bz^2 + O(z^3) \ ,$$

we find

$$t_0 - t = AH_0(t_0 - t) + AH_0^2\Big(\frac{q_0}{2} + 1\Big)(t_0 - t)^2 + BH_0^2(t_0 - t)^2 \ ,$$

which implies

$$AH_0 = 1 \ , \quad A = \frac{1}{H_0} \ ,$$

$$AH_0^2\Big(\frac{q_0}{2} + 1\Big) + BH_0^2 = 0 \ , \quad \Big(\frac{q_0}{2} + 1\Big) + BH_0 = 0 \ ,$$

$$BH_0 = -\frac{q_0}{2} - 1 \ , \quad B = -\frac{1}{H_0}\Big(\frac{q_0}{2} + 1\Big) \ .$$

Putting together, we obtain

$$t_0 - t = \frac{1}{H_0}\Big(z - \Big(\frac{q_0}{2} + 1\Big)z^2\Big) + O(z^3) \ .$$

Our final goal is to find the Hubble's law, which is written in terms of distance

$$\int_t^{t_0} \frac{dt}{a(t)} = \int_0^r \frac{dr}{\sqrt{1 - kr^2}} \simeq \int_0^r dr = r \ .$$

Therefore, we have

$$\begin{aligned}
r &\simeq \int_t^{t_0} \frac{dt}{a(t)} \simeq \int_t^{t_0} dt \ \frac{1}{a_0}(1 + H_0(t - t_0))^{-1} \simeq \int_t^{t_0} dt \ \frac{1}{a_0}(1 - H_0(t - t_0)) \\
&= \int_t^{t_0} dt \ \frac{1}{a_0}(1 + H_0 t_0 - H_0 t) = \frac{1}{a_0}(t + H_0 t_0 t - H_0 \frac{t^2}{2})\Big|_t^{t_0} \\
&= \frac{1}{a_0}(t_0 + H_0 t_0^2 - H_0 \frac{t_0^2}{2} - t - H_0 t_0 t + H_0 \frac{t^2}{2}) + O((t - t_0)^3) \\
&= \frac{1}{a_0}\Big((t_0 - t) + H_0 \frac{t_0^2}{2} - H_0 t_0 t + H_0 \frac{t^2}{2}\Big) + O((t - t_0)^3) \\
&= \frac{1}{a_0}\Big((t_0 - t) + \frac{H_0}{2}(t_0 - t)^2\Big) + O((t - t_0)^3) \ .
\end{aligned}$$

Finally, combining the two expressions, we obtain

$$r \simeq \frac{1}{H_0 a_0}\Big(z - (\frac{q_0}{2} + 1)z^2 + \frac{1}{2}z^2\Big) + O(z^3) = \frac{1}{H_0 a_0}\Big(z + \frac{1}{2}(1 - q_0)z^2\Big) + O(z^3)$$

and the distance-luminosity relation becomes

$$\begin{aligned} d_L &= a_0 r(1+z) = a_0(1+z)\frac{1}{H_0 a_0}\Big(z + \frac{1}{2}(1-q_0)z^2\Big) + O(z^3) \\ &\simeq \frac{1}{H_0}\Big(z + \frac{1}{2}(1-q_0)z^2\Big) + O(z^3) \ , \end{aligned}$$

which is indeed the so-called Hubble's law. Physically, it means that galaxies recede from us at a velocity $v \sim z$ proportional to the distance d_L.

10.4 Standard model of cosmology

The solutions of the Friedmann equations shows that the Universe must have started from a singularity at $a = 0$ and expands from that ever since. This can be seen by writing the second Friedmann equation in terms of the deceleration parameter

$$q = \frac{4\pi G_N}{3H^2}(\rho + 3p) \ .$$

For vanishing cosmological constant, we have $q > 0$ or $\ddot{a} < 0$, hence the universe is decelerating its expansion due to gravitational attraction. Furthermore, given that $a > 0$ (by definition), $\dot{a} > 0$ (red-shift) and $\ddot{a} < 0$, a does not have a turning point and it must have reached $a = 0$ in finite time. For a radiation-dominated universe, the energy density goes like $\rho \sim a^{-4}$, which menas that for $a \to 0$, it goes to infinity. This is the basis of the Hot Big Bang model. Its main adversary was the steady model, in which energy density remains constant. However, observations on the Cosmic Microwave Background radiation shows that the Universe is cooling down. The CMB radiation is the almost relic radiation generated by the decoupled of radiation from matter on the last scattering surface. Before, photons were in thermal equilibrium with electrons and protons, and the only way to understand the physics of that period is through gravitational waves. Another important observational feedbacks from the CMB are the fact that its fine anisotropies shows that the Universe is flat, but there is an high degree of isotropy. The latter is very suspiciuos, because it means that the light we receive from different direction in the sky should not have the same temperature, unless they were causally connected in the past. How to solve this problem, called the Horizon problem?

Suppose we place ourselves at $r = 0$ and $t = 0$. We want to find the proper distance that the photon has travelled from $r = r_0$ and $t = -t_0$ to us

$$R \sim a(-t_0) r_s \ .$$

Using the expression above, we need to evaluate the integral

$$r_0 = \int_0^{r_0} dr \sim \int_{-t_0}^{0} \frac{dt}{a(t)} \ .$$

Consider first a matter/radiation-dominated Universe, so that

$$a(t) \sim t^{\alpha} \ , \quad r_0 \sim \int_{-t_0}^{0} dt \ t^{-\alpha} \sim t_0^{1-\alpha} \ ,$$

where $\alpha = 2/3, 1/2$. Therefore, the proper distance is given by

$$R \sim a(-t_0) r_0 = t_0^{\alpha} t_0^{1-\alpha} = t_0 \ ,$$

which coincides with Minkowski. Furthermore,

$$\dot{a} \sim t^{\alpha-1} \ , \quad H = \frac{\dot{a}}{a} \sim t^{-1} \ ,$$

and the particle horizon grows linearly with time

$$R_H \sim \frac{1}{H(-t_0)} \sim t_0 \ . \tag{10.2}$$

However, in a dark energy-dominated Universe, we have

$$a(t) \sim \exp(H_0 t) \ , \quad r_s \sim \frac{\exp(-H_0 t)}{H_0} \ ,$$

and the particle horizon is constant

$$R \sim \exp(H_0 t_0) r_s \sim \frac{1}{H_0} \ .$$

This can explain CMB homogeneity: the Universe started so small to have all particles in causal connection and, then, there was an inflation phase of rapid expansion, that eventually ended.

We conclude this chapter with experimental results. As said before, the CMB provides us that the universe is flat with $\Omega \simeq 1$ and density $\rho_0 = \rho_{crit} \simeq 10^{-29} g/cm^3$. The contribution to density can be identified by the following sources

1. radiation is negligible with $\rho_{rad} \simeq 0$
2. massive baryonic matter with $\rho_{mat}/\rho_0 \simeq 5\%$
3. dark matter with $\rho_{DM}/\rho_0 \simeq 25\%$
4. dark energy with $\rho_{DE}/\rho_0 \simeq 70\%$

Dark matter is needed to explain the rotation of galaxies and dark energy to explain the expansion of the Universe.

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